

#1 · Musculoskeletal System

What is a Maitland Grade I joint mobilization?

Answer

A small amplitude oscillation performed at the beginning of available range, used primarily to reduce pain rather than increase range of motion.

#2 · Musculoskeletal System

What is a Maitland Grade II mobilization?

Answer

A large amplitude oscillation performed within the available range of motion but not reaching the end of range, used to reduce pain and gently improve mobility.

#3 · Musculoskeletal System

What is a Maitland Grade III mobilization?

Answer

A large amplitude oscillation that moves into tissue resistance up to the limit of available range, used to address stiffness and increase range of motion.

#4 · Musculoskeletal System

What is a Maitland Grade IV mobilization?

Answer

A small amplitude oscillation performed at the very end of available range into tissue resistance, used to increase range when stiffness rather than pain limits movement.

#5 · Musculoskeletal System

What is a Grade V mobilization also called, and how does it differ from Grades I to IV?

Answer

A high velocity low amplitude thrust manipulation taken past the physiologic limit to the anatomic limit, delivered as a fast small thrust rather than a slow oscillation.

#6 · Musculoskeletal System

In the Kaltenborn grading system, what does a Grade I traction accomplish?

Answer

It nullifies joint compression forces without actually separating the joint surfaces, used to reduce pain and is often applied before any glide or oscillation technique.

#7 · Musculoskeletal System

In the Kaltenborn system, what is the difference between Grade II and Grade III traction?

Answer

Grade II takes up the slack in the joint capsule to the first point of tissue resistance, while Grade III stretches the capsule beyond that point to increase joint mobility.

#8 · Musculoskeletal System

What is the concave-convex rule used for in joint mobilization?

Answer

It predicts the direction the convex or concave bony surface should glide relative to the direction the bone is rolling, guiding which glide direction restores or improves joint motion.

#9 · Musculoskeletal System

According to the convex-concave rule, if a convex surface moves on a fixed concave surface, which direction does the roll and glide occur?

Answer

The roll and the glide occur in opposite directions, so if the bone rolls anteriorly the convex surface glides posteriorly.

#10 · Musculoskeletal System

What special test assesses anterior cruciate ligament integrity by applying an anterior force to the tibia with the knee flexed 20 to 30 degrees?

Answer

The Lachman test, considered the most sensitive clinical test for ACL rupture.

#11 · Musculoskeletal System

What is a positive McMurray test indicative of, and how is it performed?

Answer

It indicates a meniscal tear, performed by flexing the knee fully then extending it while applying internal or external tibial rotation and valgus or varus stress, with a palpable or audible click being a positive sign.

#12 · Musculoskeletal System

What does a positive posterior drawer test suggest, and how does it differ from the anterior drawer test?

Answer

It suggests posterior cruciate ligament insufficiency, tested by pushing the tibia posteriorly on a fixed femur with the knee flexed 90 degrees, whereas the anterior drawer pulls the tibia forward to test the ACL.

#13 · Musculoskeletal System

What is the Neer impingement test and what does a positive result suggest?

Answer

The examiner passively flexes the shoulder forward while stabilizing the scapula, and reproduction of pain suggests subacromial impingement of the supraspinatus tendon or subacromial bursa.

#14 · Musculoskeletal System

How is the Hawkins-Kennedy test performed and what does it assess?

Answer

The shoulder and elbow are flexed to 90 degrees and the examiner internally rotates the shoulder, and reproduced pain suggests subacromial impingement.

#15 · Musculoskeletal System

What does a positive empty can test (Jobe test) indicate?

Answer

Weakness or pain with resisted shoulder elevation in the scapular plane with the thumb pointing down suggests supraspinatus tendon pathology or tear.

#16 · Musculoskeletal System

What is the Speed's test used to assess, and what is a positive finding?

Answer

It assesses biceps tendon pathology; the examiner resists shoulder flexion with the elbow extended and forearm supinated, and pain in the bicipital groove is a positive result.

#17 · Musculoskeletal System

What does the Yergason test assess and how is it performed?

Answer

It assesses biceps tendon stability in the bicipital groove; the examiner resists forearm supination while the elbow is flexed to 90 degrees, and pain or a popping sensation indicates a positive test.

#18 · Musculoskeletal System

What does a positive FABER (Patrick) test suggest?

Answer

Pain in the groin suggests hip joint pathology, while pain in the posterior sacroiliac region suggests sacroiliac joint dysfunction, tested with the hip flexed, abducted, and externally rotated.

#19 · Musculoskeletal System

What does the Ober test assess and what indicates a positive result?

Answer

It assesses iliotibial band tightness; with the patient side-lying and the test leg abducted and extended at the hip, failure of the leg to adduct toward the table when released indicates tightness.

#20 · Musculoskeletal System

What is the Thomas test used for and what does a positive result indicate?

Answer

It assesses hip flexor tightness; with one hip pulled to the chest, if the opposite leg lifts off the table or the knee extends, this indicates a tight iliopsoas or rectus femoris.

#21 · Musculoskeletal System

What does a positive Phalen test suggest and how is it performed?

Answer

It suggests carpal tunnel syndrome, tested by holding both wrists in maximal flexion with the dorsal surfaces touching for 60 seconds, with reproduction of numbness or tingling in the median nerve distribution being positive.

#22 · Musculoskeletal System

What is a positive Tinel sign at the wrist, and what does it indicate?

Answer

Tapping over the median nerve at the carpal tunnel reproduces tingling or an electric shock sensation into the thumb, index, and middle fingers, indicating carpal tunnel syndrome.

#23 · Musculoskeletal System

What does a positive Spurling test suggest and how is it performed?

Answer

It suggests cervical nerve root compression; the examiner extends and laterally flexes the neck toward the affected side while applying gentle downward pressure, and reproduction of radicular symptoms is positive.

#24 · Musculoskeletal System

What is the straight leg raise test used to assess and what angle range typically reproduces true sciatic symptoms?

Answer

It assesses lumbar nerve root irritation or disc herniation, and reproduction of radicular leg pain between 30 and 70 degrees of hip flexion is considered a positive finding.

#25 · Musculoskeletal System

What does the slump test add to a standard straight leg raise assessment?

Answer

It combines cervical, thoracic, and lumbar flexion with knee extension and ankle dorsiflexion to increase neural tension throughout the entire spinal cord and dural sleeve, increasing sensitivity for neural tissue involvement.

#26 · Musculoskeletal System

What are the four stages of fracture healing in order?

Answer

Hematoma formation, soft (fibrocartilaginous) callus formation, hard (bony) callus formation, and bone remodeling.

#27 · Musculoskeletal System

Approximately how long does the hematoma formation stage of fracture healing last?

Answer

It occurs within the first few hours to about 1 week after the fracture, as a blood clot forms at the fracture site and inflammatory cells begin to arrive.

#28 · Musculoskeletal System

During which stage of fracture healing does fibrocartilage bridge the fracture gap, and roughly when does it occur?

Answer

During the soft callus stage, occurring roughly from 2 to 3 weeks post-fracture, providing initial but not yet weight-bearing-stable stability.

#29 · Musculoskeletal System

When does hard callus formation typically occur and what happens during it?

Answer

It typically occurs from about 3 to 4 months post-fracture, as osteoblasts convert the soft callus into woven bone, providing enough stability for progressive weight bearing.

#30 · Musculoskeletal System

What is Wolff's law and how does it apply to fracture remodeling?

Answer

Wolff's law states that bone remodels and reorients its trabecular structure in response to the mechanical stresses placed upon it, so controlled progressive loading during remodeling helps restore normal bone architecture and strength.

#31 · Musculoskeletal System

What are standard total hip arthroplasty posterior approach precautions?

Answer

Avoid hip flexion beyond 90 degrees, avoid hip adduction past midline, and avoid combined hip internal rotation, since these motions risk posterior dislocation.

#32 · Musculoskeletal System

How do anterior approach total hip arthroplasty precautions differ from posterior approach precautions?

Answer

Anterior approach precautions restrict hip extension, external rotation, and adduction, the opposite motions restricted in a posterior approach, because the risk of dislocation is directed anteriorly instead of posteriorly.

#33 · Musculoskeletal System

What is the typical weight-bearing status and early ROM goal in the first 1 to 2 weeks after a standard total knee arthroplasty?

Answer

Weight bearing as tolerated with an assistive device is typical, with early goals focused on achieving knee extension to 0 degrees and flexion progressing toward 90 degrees to prevent contracture.

#34 · Musculoskeletal System

What is the general timeline and criteria for beginning running progression after ACL reconstruction?

Answer

Running typically begins around 3 to 4 months post-surgery once quadriceps strength reaches roughly 70 to 80 percent of the uninvolved side and the patient demonstrates good single leg control without pain or effusion.

#35 · Musculoskeletal System

Why are open kinetic chain quadriceps exercises typically restricted in early ACL reconstruction rehab, especially with a patellar tendon or hamstring graft?

Answer

Open chain knee extension from about 0 to 45 degrees places high anterior shear stress on the healing graft, so this arc is often avoided or limited in the first 4 to 6 weeks depending on graft type and surgeon protocol.

#36 · Musculoskeletal System

What return-to-sport criteria are commonly used after ACL reconstruction beyond simple time since surgery?

Answer

Quadriceps strength within 90 percent limb symmetry index, single leg hop test symmetry above 90 percent, and full pain-free range of motion are commonly required in addition to a minimum time frame of 6 to 9 months.

#37 · Musculoskeletal System

What movement restrictions are typical during the first 4 to 6 weeks after a rotator cuff repair?

Answer

Active range of motion and active-assisted stretching beyond protected limits are restricted, with the arm kept in a sling and passive range of motion only performed within surgeon-specified limits to protect the healing repair.

#38 · Musculoskeletal System

When does active range of motion typically begin after a rotator cuff repair, and why is this delayed?

Answer

Active range of motion typically begins around 6 weeks post-surgery, delayed because tendon-to-bone healing requires this time before the repaired tendon can tolerate active muscular contraction without risking re-tear.

#39 · Musculoskeletal System

What are standard log-roll precautions after lumbar spinal fusion?

Answer

The patient rolls the trunk as one unit while keeping the spine in neutral alignment, avoiding independent twisting or bending of the trunk during bed mobility to protect the fusion site.

#40 · Musculoskeletal System

What activities are typically restricted after spinal fusion under the acronym BLT?

Answer

Bending, Lifting, and Twisting are restricted, since these movements place shear and rotational stress on the healing fusion segment.

#41 · Musculoskeletal System

What is normal shoulder flexion range of motion measured goniometrically?

Answer

Approximately 0 to 180 degrees.

#42 · Musculoskeletal System

What is normal shoulder abduction range of motion?

Answer

Approximately 0 to 180 degrees.

#43 · Musculoskeletal System

What is normal shoulder external rotation and internal rotation range of motion?

Answer

External rotation is approximately 0 to 90 degrees and internal rotation is approximately 0 to 70 degrees, measured with the shoulder abducted to 90 degrees.

#44 · Musculoskeletal System

What is normal elbow flexion and extension range of motion?

Answer

Elbow flexion is approximately 0 to 150 degrees, and extension is 0 degrees, with some individuals having up to 10 degrees of hyperextension considered normal.

#45 · Musculoskeletal System

What is normal forearm pronation and supination range of motion?

Answer

Both pronation and supination are approximately 0 to 80 to 90 degrees each.

#46 · Musculoskeletal System

What is normal wrist flexion and extension range of motion?

Answer

Wrist flexion is approximately 0 to 80 degrees and wrist extension is approximately 0 to 70 degrees.

#47 · Musculoskeletal System

What is normal hip flexion range of motion measured with the knee flexed versus extended?

Answer

Hip flexion with the knee flexed is approximately 0 to 120 degrees, while with the knee extended it is somewhat less, around 0 to 90 to 100 degrees, due to hamstring tension.

#48 · Musculoskeletal System

What is normal hip abduction and adduction range of motion?

Answer

Hip abduction is approximately 0 to 45 degrees and hip adduction is approximately 0 to 30 degrees.

#49 · Musculoskeletal System

What is normal knee flexion range of motion?

Answer

Approximately 0 to 135 to 150 degrees.

#50 · Musculoskeletal System

What is normal ankle dorsiflexion and plantarflexion range of motion?

Answer

Dorsiflexion is approximately 0 to 20 degrees and plantarflexion is approximately 0 to 50 degrees.

#51 · Musculoskeletal System

What is normal cervical spine rotation range of motion?

Answer

Approximately 0 to 80 degrees to each side.

#52 · Musculoskeletal System

What is normal lumbar spine flexion range of motion?

Answer

Approximately 0 to 60 degrees, though functional lumbar flexion is often measured using tape measure methods like the modified Schober test rather than a goniometer.

#53 · Musculoskeletal System

What does a Manual Muscle Test grade of 5/5 (Normal) indicate?

Answer

The patient completes full available range of motion against gravity and tolerates maximal resistance from the examiner.

#54 · Musculoskeletal System

What does a Manual Muscle Test grade of 4/5 (Good) indicate?

Answer

The patient completes full range of motion against gravity and tolerates moderate resistance but breaks with strong resistance from the examiner.

#55 · Musculoskeletal System

What does a Manual Muscle Test grade of 3/5 (Fair) indicate?

Answer

The patient completes full available range of motion against gravity but cannot tolerate any additional manual resistance.

#56 · Musculoskeletal System

What does a Manual Muscle Test grade of 2/5 (Poor) indicate?

Answer

The patient completes full range of motion only with gravity eliminated, meaning the limb moves in a gravity-minimized plane rather than against gravity.

#57 · Musculoskeletal System

What does a Manual Muscle Test grade of 1/5 (Trace) indicate?

Answer

A palpable or visible muscle contraction is present but produces no actual joint movement.

#58 · Musculoskeletal System

What does a Manual Muscle Test grade of 0/5 indicate?

Answer

No palpable or observable muscle contraction is present at all.

#59 · Musculoskeletal System

What does adding a plus or minus to a Manual Muscle Test grade, such as 4+ or 3-, signify?

Answer

A plus indicates the patient tolerated slightly more resistance or range than the base grade describes, while a minus indicates slightly less, allowing finer grading distinctions between the whole-number grades.

#60 · Musculoskeletal System

How does osteoarthritis typically present differently from rheumatoid arthritis in terms of joint symptom pattern?

Answer

Osteoarthritis typically presents asymmetrically in weight-bearing joints with pain that worsens with activity and improves with rest, while rheumatoid arthritis presents symmetrically, often in smaller joints, with prolonged morning stiffness lasting over an hour.

#61 · Musculoskeletal System

What is the underlying pathology of osteoarthritis?

Answer

Progressive degeneration and loss of articular cartilage with subsequent bony changes such as osteophyte formation and subchondral sclerosis, driven primarily by mechanical wear rather than systemic inflammation.

#62 · Musculoskeletal System

What is the underlying pathology of rheumatoid arthritis?

Answer

A chronic systemic autoimmune disease in which the immune system attacks the synovial lining of joints, causing synovitis, pannus formation, and eventual erosion of cartilage and bone.

#63 · Musculoskeletal System

What deformities are classically associated with rheumatoid arthritis of the hand?

Answer

Ulnar deviation of the fingers at the metacarpophalangeal joints, swan neck deformity, and boutonniere deformity.

#64 · Musculoskeletal System

What are Heberden's nodes and Bouchard's nodes, and with which condition are they associated?

Answer

Heberden's nodes are bony enlargements at the distal interphalangeal joints and Bouchard's nodes occur at the proximal interphalangeal joints, both classically associated with osteoarthritis rather than rheumatoid arthritis.

#65 · Musculoskeletal System

What precautions should a therapist take when treating a patient with rheumatoid arthritis during an active disease flare?

Answer

Avoid aggressive stretching or resistive exercise to inflamed joints, use gentle range of motion and isometric exercise instead, and monitor for cervical spine instability such as atlantoaxial subluxation before performing neck movements.

#66 · Musculoskeletal System

What is the pathophysiology of tendinopathy, and why is the older term tendinitis considered less accurate for chronic cases?

Answer

Chronic tendinopathy involves collagen disorganization and degeneration with minimal true inflammatory cells present, so the term tendinosis or tendinopathy is preferred over tendinitis, which implies an inflammatory process that is often absent in chronic presentations.

#67 · Musculoskeletal System

What is the current evidence-based first-line exercise intervention for Achilles tendinopathy?

Answer

Progressive eccentric loading exercises, such as the Alfredson heel-drop protocol, performed over the edge of a step with the knee both straight and bent.

#68 · Musculoskeletal System

What is the common site and cause of lateral epicondylitis?

Answer

Degeneration of the extensor carpi radialis brevis tendon at its origin on the lateral epicondyle, typically caused by repetitive wrist extension and gripping activities.

#69 · Musculoskeletal System

What test is used to assess lateral epicondylitis and how is it performed?

Answer

The Cozen's test, performed by resisting wrist extension with the elbow extended and forearm pronated, with reproduction of pain at the lateral epicondyle being a positive finding.

#70 · Musculoskeletal System

What structure is involved in patellar tendinopathy, and what population is it commonly seen in?

Answer

The patellar tendon at its origin on the inferior pole of the patella, commonly seen in athletes performing repetitive jumping activities such as basketball and volleyball, hence its nickname 'jumper's knee.'

#71 · Musculoskeletal System

What are the main subgroups in the treatment-based classification system for low back pain?

Answer

Manipulation, stabilization, specific exercise (which includes extension, flexion, and lateral shift subgroups), and traction.

#72 · Musculoskeletal System

What clinical prediction rule findings suggest a patient with low back pain is likely to respond well to spinal manipulation?

Answer

Symptom duration less than 16 days, no symptoms distal to the knee, low fear-avoidance beliefs, at least one hip with more than 35 degrees of internal rotation, and hypomobility of the lumbar spine.

#73 · Musculoskeletal System

In the McKenzie method, what does directional preference mean and what is the classic example for a posterior disc derangement?

Answer

Directional preference is the specific repeated movement direction that centralizes or reduces a patient's symptoms; for a posterior disc derangement this is typically repeated lumbar extension, which centralizes leg pain back toward the spine.

#74 · Musculoskeletal System

What is centralization of symptoms and why is it clinically significant in low back pain assessment?

Answer

Centralization is the phenomenon of distal or peripheral symptoms retreating toward the midline of the spine with repeated movement testing, and it is a favorable prognostic sign associated with disc-related pathology that often responds well to conservative treatment.

#75 · Musculoskeletal System

What are red flags that should prompt referral rather than continued conservative treatment for low back pain?

Answer

Unexplained weight loss, history of cancer, progressive neurologic deficit, saddle anesthesia, bowel or bladder dysfunction, fever, and significant trauma, since these may indicate cauda equina syndrome, fracture, infection, or malignancy.

#76 · Musculoskeletal System

What are the eight phases of the normal gait cycle?

Answer

Initial contact, loading response, midstance, terminal stance, pre-swing, initial swing, midswing, and terminal swing.

#77 · Musculoskeletal System

What percentage of the gait cycle is spent in stance phase versus swing phase during normal walking?

Answer

Stance phase accounts for approximately 60 percent of the gait cycle and swing phase accounts for approximately 40 percent.

#78 · Musculoskeletal System

What is a Trendelenburg gait and what muscle weakness causes it?

Answer

A gait pattern in which the pelvis drops on the swing-leg side during single-limb stance, caused by weakness of the hip abductors, primarily the gluteus medius, on the stance leg side.

#79 · Musculoskeletal System

What is a steppage gait and what causes it?

Answer

A gait pattern with excessive hip and knee flexion during swing phase to clear the foot off the ground, caused by weakness or paralysis of the ankle dorsiflexors, commonly seen with foot drop from common peroneal nerve injury.

#80 · Musculoskeletal System

What is an antalgic gait?

Answer

A pain-avoidance gait pattern characterized by a shortened stance phase on the painful limb to minimize weight-bearing time and discomfort.

#81 · Musculoskeletal System

What gait deviation is commonly seen with quadriceps weakness during stance phase?

Answer

The patient may lean the trunk forward or push the knee into hyperextension (back-kneeing) to stabilize the knee and substitute for insufficient quadriceps control.

#82 · Musculoskeletal System

What is a scissoring gait pattern and with what condition is it commonly associated?

Answer

A gait pattern with excessive hip adduction causing the knees and thighs to cross the midline during swing, commonly seen in spastic cerebral palsy or other upper motor neuron conditions with hip adductor spasticity.

#83 · Musculoskeletal System

What is genu recurvatum and how might it be observed during gait?

Answer

Genu recurvatum is hyperextension of the knee beyond normal limits, which during gait may be seen as excessive knee hyperextension at midstance, often used to compensate for quadriceps weakness or plantarflexor tightness.

#84 · Musculoskeletal System

What is the normal alignment of the sagittal plane spinal curves from superior to inferior?

Answer

Cervical lordosis, thoracic kyphosis, lumbar lordosis, and sacral kyphosis.

#85 · Musculoskeletal System

What is kyphoscoliosis?

Answer

A postural or structural deformity combining excessive thoracic kyphosis in the sagittal plane with lateral curvature and rotation of the spine (scoliosis) in the frontal and transverse planes.

#86 · Musculoskeletal System

What test is used to differentiate structural scoliosis from functional (postural) scoliosis?

Answer

The Adam's forward bend test; a rib hump or rotational prominence that persists during forward bending suggests structural scoliosis, while a curve that disappears with forward bending suggests functional or postural scoliosis.

#87 · Musculoskeletal System

What is upper crossed syndrome and what muscle imbalances characterize it?

Answer

A postural pattern with tightness of the upper trapezius and pectoralis muscles combined with weakness of the deep neck flexors and lower/middle trapezius, resulting in a forward head, rounded shoulders posture.

#88 · Musculoskeletal System

What is lower crossed syndrome and what muscle imbalances characterize it?

Answer

A postural pattern with tightness of the hip flexors and lumbar erector spinae combined with weakness of the abdominal muscles and gluteal muscles, resulting in an anterior pelvic tilt and increased lumbar lordosis.

#89 · Musculoskeletal System

What is a flatback posture and what spinal curve is affected?

Answer

A posture characterized by a decreased or flattened lumbar lordosis, often accompanied by a posterior pelvic tilt, which can limit shock absorption and increase stress on lumbar discs during activity.

#90 · Musculoskeletal System

What is the correct order of therapeutic exercise progression from least to most demanding for contraction type?

Answer

Isometric exercise progresses to isotonic (concentric and eccentric) exercise, which then progresses to isokinetic or plyometric exercise as tolerance and control improve.

#91 · Musculoskeletal System

What is the general principle behind progressing from closed kinetic chain to open kinetic chain exercise, or vice versa, in rehabilitation?

Answer

Closed kinetic chain exercises, where the distal segment is fixed, generally provide greater joint compression and proprioceptive feedback with less shear force, so they are often introduced earlier for joint protection, while open kinetic chain exercises isolate specific muscles and are added later for targeted strengthening.

#92 · Musculoskeletal System

What is the SAID principle and how does it guide exercise prescription?

Answer

SAID stands for Specific Adaptation to Imposed Demands, meaning the body adapts specifically to the type, intensity, and pattern of stress placed upon it, so exercise should mimic the functional demands the patient will ultimately need to perform.

#93 · Musculoskeletal System

What is plyometric exercise and what physiological principle underlies its effectiveness?

Answer

Plyometric exercise uses a rapid eccentric muscle lengthening (loading phase) immediately followed by a rapid concentric contraction (unloading phase), exploiting the stretch-shortening cycle to produce a more powerful subsequent contraction.

#94 · Musculoskeletal System

What is the general order of load progression recommended when advancing a strengthening program using the overload principle?

Answer

Increase repetitions first, then increase resistance or intensity, and finally increase the complexity or velocity of the movement, adjusting only one variable at a time to safely monitor tolerance.

#95 · Musculoskeletal System

What is delayed onset muscle soreness and when does it typically peak?

Answer

Delayed onset muscle soreness is pain and stiffness resulting from microtrauma to muscle fibers, particularly after unaccustomed eccentric exercise, typically peaking 24 to 72 hours after the activity.

#96 · Musculoskeletal System

What is the normal end feel of elbow extension and what does this end feel classification mean?

Answer

A bony (hard) end feel, meaning two hard surfaces (the olecranon process and the olecranon fossa) come into contact, producing an abrupt stop with no give.

#97 · Musculoskeletal System

What is a capsular end feel and give an example of a joint motion that normally has one?

Answer

A firm end feel with some give, similar to stretching a piece of leather, normally present at the end of shoulder external rotation due to tension in the anterior joint capsule.

#98 · Musculoskeletal System

What does an empty end feel indicate during passive range of motion testing?

Answer

The examiner encounters no mechanical resistance to further motion, but the patient stops the motion due to pain, often indicating an acute inflammatory condition, bursitis, or a serious pathology such as a tumor or fracture.

#99 · Musculoskeletal System

What is the normal capsular pattern of restriction for the glenohumeral joint?

Answer

External rotation is most limited, followed by abduction, followed by internal rotation, reflecting a proportional pattern of capsular tightness.

#100 · Musculoskeletal System

What is the capsular pattern of restriction for the hip joint?

Answer

Flexion and internal rotation are most limited, followed by abduction, with relatively little restriction to extension and external rotation.

#101 · Musculoskeletal System

What is the capsular pattern of restriction for the knee joint?

Answer

Flexion is more limited than extension, a proportional pattern consistent with true capsular joint restriction such as osteoarthritis.

#102 · Musculoskeletal System

What is adhesive capsulitis, and what are its three clinical stages?

Answer

Adhesive capsulitis, or frozen shoulder, is a condition of progressive capsular fibrosis and thickening; its stages are the freezing (painful) stage, the frozen (stiffening) stage, and the thawing (recovery) stage.

#103 · Musculoskeletal System

What is the difference between a Grade I and Grade II ligament sprain?

Answer

A Grade I sprain involves mild stretching of ligament fibers with minimal microscopic tearing and no joint instability, while a Grade II sprain involves partial tearing of the ligament with some laxity but the joint remains grossly stable.

#104 · Musculoskeletal System

What defines a Grade III ligament sprain?

Answer

A complete tear or rupture of the ligament, resulting in gross joint instability on stress testing.

#105 · Musculoskeletal System

How are muscle strains graded from Grade I to Grade III?

Answer

Grade I is a mild strain with minimal fiber disruption and mild pain, Grade II is a partial tear with moderate pain and some loss of strength, and Grade III is a complete rupture of the muscle or its tendon with significant functional loss.

#106 · Musculoskeletal System

What is the RICE protocol used for acute musculoskeletal injury management, and what is the modern update to this approach?

Answer

RICE stands for Rest, Ice, Compression, and Elevation; the modern updated approach, POLICE, replaces complete rest with Protection and Optimal Loading to encourage earlier controlled movement for better tissue healing.

#107 · Musculoskeletal System

What is a Colles' fracture and what deformity does it classically produce?

Answer

A distal radius fracture typically caused by a fall on an outstretched hand, producing a classic dorsally displaced 'dinner fork' deformity of the wrist.

#108 · Musculoskeletal System

What is the significance of a scaphoid fracture and why is it clinically important to recognize?

Answer

The scaphoid has a precarious blood supply entering distally, so a fracture, especially of the proximal pole, carries a high risk of avascular necrosis and nonunion if missed or inadequately immobilized.

#109 · Musculoskeletal System

What special test is used to screen for a scaphoid fracture and what physical finding suggests one?

Answer

Tenderness in the anatomical snuffbox on palpation, along with pain with axial loading of the thumb, are classic clinical signs suggestive of a scaphoid fracture.

#110 · Musculoskeletal System

What is a hip fracture classification distinction between intracapsular and extracapsular fractures, and why does it matter?

Answer

Intracapsular fractures occur within the joint capsule (femoral neck) and carry a higher risk of avascular necrosis due to disrupted blood supply, while extracapsular fractures (intertrochanteric or subtrochanteric) occur outside the capsule and generally have better blood supply and healing potential.

#111 · Musculoskeletal System

What weight-bearing status is typical after an open reduction internal fixation for a hip fracture, and why might this differ from an arthroplasty?

Answer

Weight bearing as tolerated is commonly allowed after arthroplasty since the implant provides immediate structural stability, whereas internal fixation of certain fracture patterns may require touchdown or partial weight bearing initially to protect the healing bone until adequate callus forms.

#112 · Musculoskeletal System

What is the difference between a stress fracture and a traumatic fracture?

Answer

A stress fracture results from repetitive submaximal loading over time that exceeds the bone's ability to remodel and repair, while a traumatic fracture results from a single acute overwhelming force.

#113 · Musculoskeletal System

What are common risk factors for a female athlete to develop a stress fracture, referencing the Female Athlete Triad?

Answer

Low energy availability or disordered eating, menstrual dysfunction (amenorrhea), and decreased bone mineral density together increase stress fracture risk in this population.

#114 · Musculoskeletal System

What is spondylolysis and how does it differ from spondylolisthesis?

Answer

Spondylolysis is a stress fracture or defect of the pars interarticularis of a vertebra, while spondylolisthesis is the forward slippage of one vertebral body relative to the one below it, which can develop as a consequence of bilateral spondylolysis.

#115 · Musculoskeletal System

What test can help identify spondylolysis or spondylolisthesis clinically?

Answer

The stork test (single leg lumbar extension test), in which the patient stands on one leg and extends the lumbar spine; reproduction of unilateral low back pain suggests a pars defect.

#116 · Musculoskeletal System

What is piriformis syndrome and what nerve is commonly implicated?

Answer

A condition where the piriformis muscle compresses or irritates the sciatic nerve as it passes near or through the muscle, producing buttock pain that may radiate down the posterior thigh.

#117 · Musculoskeletal System

What test is commonly used to assess for piriformis syndrome?

Answer

The FAIR test (Flexion, Adduction, and Internal Rotation of the hip), which stretches and compresses the piriformis against the sciatic nerve, reproducing symptoms if the piriformis is the source.

#118 · Musculoskeletal System

What is patellofemoral pain syndrome and what factors commonly contribute to it?

Answer

Anterior knee pain related to abnormal patellar tracking within the femoral trochlear groove, commonly contributed to by quadriceps weakness (especially the vastus medialis obliquus), hip abductor and external rotator weakness, and iliotibial band tightness.

#119 · Musculoskeletal System

What is the clarke sign (patellar grind test) used to assess?

Answer

It assesses patellofemoral joint pain by applying gentle downward pressure on the superior patella while the patient contracts the quadriceps; reproduction of pain or grinding suggests patellofemoral joint dysfunction, though its diagnostic value is debated.

#120 · Musculoskeletal System

What is the difference between grade 1, 2, and 3 acromioclavicular joint sprains?

Answer

Grade 1 involves a sprain of the acromioclavicular ligaments with no instability, Grade 2 involves a rupture of the acromioclavicular ligaments with partial instability and some elevation of the clavicle, and Grade 3 involves complete rupture of both the acromioclavicular and coracoclavicular ligaments with gross instability and a visible step deformity.

#121 · Musculoskeletal System

What is impingement syndrome of the shoulder and what structures are typically compressed?

Answer

Compression of the supraspinatus tendon, subacromial bursa, and long head of the biceps tendon between the humeral head and the coracoacromial arch, often occurring during overhead arm elevation.

#122 · Musculoskeletal System

What is the drop arm test and what does a positive result indicate?

Answer

The patient's arm is passively abducted to 90 degrees and asked to slowly lower it; inability to control the descent or a sudden drop suggests a full-thickness rotator cuff tear, most often of the supraspinatus.

#123 · Musculoskeletal System

What is scapular dyskinesis and how might it be observed clinically?

Answer

Abnormal scapular positioning or movement pattern during arm elevation, often observed as winging, premature scapular elevation, or asymmetric scapular rhythm compared to the contralateral side.

#124 · Musculoskeletal System

What causes a classic winged scapula deformity, and what test helps identify it?

Answer

Weakness or paralysis of the serratus anterior, usually from long thoracic nerve injury, causes medial winging; it is commonly identified by having the patient perform a wall push-up, which accentuates the winging.

#125 · Musculoskeletal System

What is degenerative disc disease and how does it typically affect disc height and hydration over time?

Answer

A progressive age-related process in which intervertebral discs lose water content and proteoglycan concentration, leading to decreased disc height, reduced shock-absorbing capacity, and increased stress on adjacent facet joints.

#126 · Musculoskeletal System

What is spinal stenosis and what body position typically relieves symptoms?

Answer

Spinal stenosis is a narrowing of the spinal canal or neural foramina that compresses the spinal cord or nerve roots; flexed postures, such as leaning forward or sitting, typically relieve symptoms by increasing the available space in the canal, while extension typically worsens symptoms.

#127 · Musculoskeletal System

What is neurogenic claudication and how does it differ from vascular claudication?

Answer

Neurogenic claudication is leg pain or fatigue from lumbar spinal stenosis that improves with spinal flexion or sitting even without stopping walking, whereas vascular claudication is leg pain from arterial insufficiency that improves simply with rest regardless of spinal position.

#128 · Musculoskeletal System

What is the significance of a positive bicycle test (van Gelderen test) in differentiating lumbar spinal stenosis from vascular claudication?

Answer

A patient with neurogenic claudication from spinal stenosis can typically pedal longer while leaning forward in a flexed spine position than while sitting upright, since flexion decreases neural compression, a distinction not seen with pure vascular claudication.

#129 · Musculoskeletal System

What is cauda equina syndrome and why is it considered a medical emergency?

Answer

Compression of the lumbosacral nerve roots below the level of the spinal cord, presenting with saddle anesthesia, bowel or bladder dysfunction, and bilateral leg weakness; it requires emergency surgical decompression to prevent permanent neurological damage.

#130 · Musculoskeletal System

What is the general purpose of the Oswestry Disability Index in low back pain assessment?

Answer

It is a patient-reported outcome measure that quantifies the degree of functional disability related to low back pain across activities such as personal care, lifting, walking, sitting, and standing, used to track change over time.

#131 · Musculoskeletal System

What is the difference between primary and secondary bone healing?

Answer

Primary bone healing occurs through direct osteonal remodeling without callus formation, typically requiring rigid fixation with anatomic reduction, while secondary bone healing occurs through the callus formation process and is more common with casting or less rigid fixation methods.

#132 · Musculoskeletal System

What is heterotopic ossification and in which patient population is it commonly seen after musculoskeletal trauma or surgery?

Answer

Abnormal bone formation within soft tissue outside the skeleton, commonly seen after total hip arthroplasty, traumatic brain injury, or spinal cord injury, presenting as decreased range of motion, swelling, and warmth around the affected joint.

#133 · Musculoskeletal System

What is complex regional pain syndrome and what are its hallmark clinical features?

Answer

A chronic pain condition typically following trauma or surgery, characterized by disproportionate pain, along with vasomotor changes such as skin color and temperature changes, sudomotor changes such as sweating abnormalities, edema, and eventual trophic changes to skin, hair, and nails.

#134 · Musculoskeletal System

What is the difference between active insufficiency and passive insufficiency of a two-joint muscle?

Answer

Active insufficiency occurs when a muscle is shortened over both joints it crosses and cannot generate additional effective tension, while passive insufficiency occurs when a muscle is lengthened over both joints simultaneously and cannot stretch further without limiting the available range of motion.

#135 · Musculoskeletal System

Give a classic example of active insufficiency involving the finger flexors.

Answer

Making a fist with the wrist flexed produces a weaker grip than making a fist with the wrist in a neutral or slightly extended position, because the finger flexors are actively insufficient when shortened across both the wrist and finger joints simultaneously.

#136 · Musculoskeletal System

What is the length-tension relationship of skeletal muscle and how does it relate to optimal force production?

Answer

Muscle force production is greatest at an intermediate resting length that allows optimal actin-myosin cross-bridge overlap, with force declining when the muscle is either significantly shortened or significantly lengthened beyond this optimal range.

#137 · Musculoskeletal System

What is the difference between a concentric and an eccentric muscle contraction?

Answer

A concentric contraction involves the muscle shortening while generating force to overcome resistance, while an eccentric contraction involves the muscle lengthening under tension while controlling or resisting an external load.

#138 · Musculoskeletal System

Why can muscles generally produce more force during eccentric contractions than concentric contractions at the same velocity?

Answer

During eccentric contraction, external force stretches actively cycling cross-bridges, and the passive elastic components of the muscle contribute additional resistive force, allowing greater overall force production than concentric shortening can generate.

#139 · Musculoskeletal System

What is the significance of the modified Schober test in a musculoskeletal spine examination?

Answer

It is a tape measure technique used to objectively quantify lumbar flexion range of motion by marking two points along the spine and measuring the increase in distance between them during forward bending, commonly used to assess conditions like ankylosing spondylitis.

#140 · Musculoskeletal System

What is the general goal of a closed kinetic chain versus open kinetic chain exercise for rotator cuff rehabilitation, and why might closed chain be introduced earlier post-surgically?

Answer

Closed kinetic chain exercises, such as weight-bearing on a stable surface, promote co-contraction of the rotator cuff and scapular stabilizers with lower shear stress on the healing repair, so they are often introduced earlier, while open chain exercises isolate individual rotator cuff muscles with greater joint shear and are added later in rehabilitation.

#141 · Neuromuscular and Nervous System

What is Brunnstrom Stage 1 of stroke recovery?

Answer

Flaccidity with no volitional movement present in the affected limb.

#142 · Neuromuscular and Nervous System

What characterizes Brunnstrom Stage 2?

Answer

Spasticity begins to appear and minimal voluntary movement emerges only within synergy patterns.

#143 · Neuromuscular and Nervous System

What characterizes Brunnstrom Stage 3?

Answer

Spasticity peaks and the patient can voluntarily move through the full synergy pattern, though not outside of it.

#144 · Neuromuscular and Nervous System

What characterizes Brunnstrom Stage 4?

Answer

Spasticity begins to decrease and some movement combinations that deviate from the basic synergies appear.

#145 · Neuromuscular and Nervous System

What characterizes Brunnstrom Stage 5?

Answer

Spasticity continues to decline and more difficult movement combinations independent of synergy are possible.

#146 · Neuromuscular and Nervous System

What characterizes Brunnstrom Stage 6?

Answer

Spasticity is essentially absent and isolated joint movements can be performed with near normal coordination.

#147 · Neuromuscular and Nervous System

What is the upper extremity flexor synergy pattern after stroke?

Answer

Scapular retraction and elevation, shoulder abduction and external rotation, elbow flexion, forearm supination, and wrist and finger flexion.

#148 · Neuromuscular and Nervous System

What is the upper extremity extensor synergy pattern after stroke?

Answer

Scapular protraction, shoulder adduction and internal rotation, elbow extension, forearm pronation, and wrist and finger flexion.

#149 · Neuromuscular and Nervous System

What is the lower extremity flexor synergy pattern after stroke?

Answer

Hip flexion, abduction, and external rotation, knee flexion, and ankle dorsiflexion with inversion.

#150 · Neuromuscular and Nervous System

What is the lower extremity extensor synergy pattern after stroke?

Answer

Hip extension, adduction, and internal rotation, knee extension, and ankle plantarflexion with inversion.

#151 · Neuromuscular and Nervous System

What describes a hemiplegic circumduction gait pattern?

Answer

The affected leg swings outward in an arc rather than flexing normally, typically due to inadequate knee and hip flexion or foot drop compensation.

#152 · Neuromuscular and Nervous System

What causes footdrop and steppage gait in hemiplegia?

Answer

Weak or absent ankle dorsiflexors cause the foot to drop, so the patient compensates with excessive hip and knee flexion to clear the foot during swing.

#153 · Neuromuscular and Nervous System

What is genu recurvatum during hemiplegic stance phase caused by?

Answer

Quadriceps weakness combined with plantarflexor spasticity or soleus weakness, causing the knee to hyperextend for stability during stance.

#154 · Neuromuscular and Nervous System

What is pusher syndrome after stroke?

Answer

A condition, most often from right hemisphere damage, in which the patient actively pushes toward the hemiplegic side and resists correction toward midline.

#155 · Neuromuscular and Nervous System

Which stroke syndrome typically produces contralateral leg weakness greater than arm weakness?

Answer

Anterior cerebral artery stroke, because the ACA supplies the medial frontal and parietal cortex controlling the leg.

#156 · Neuromuscular and Nervous System

Which stroke syndrome produces contralateral face and arm weakness greater than leg weakness?

Answer

Middle cerebral artery stroke, since the MCA supplies the lateral cortex representing the face and upper extremity most heavily.

#157 · Neuromuscular and Nervous System

What deficits are classic for posterior cerebral artery stroke?

Answer

Contralateral homonymous hemianopsia and visual agnosia, since the PCA supplies the occipital lobe and visual cortex.

#158 · Neuromuscular and Nervous System

What is Wallenberg syndrome and its typical cause?

Answer

Lateral medullary syndrome from posterior inferior cerebellar artery occlusion, presenting with ipsilateral facial sensory loss, dysphagia, ataxia, Horner syndrome, and contralateral body sensory loss.

#159 · Neuromuscular and Nervous System

What is the difference between Broca's and Wernicke's aphasia?

Answer

Broca's aphasia is nonfluent with intact comprehension, while Wernicke's aphasia is fluent but nonsensical with impaired comprehension.

#160 · Neuromuscular and Nervous System

What is unilateral neglect and which hemisphere lesion most commonly causes it?

Answer

Unilateral neglect is failure to attend to or respond to stimuli on one side of space, most commonly the left side following a right parietal lobe lesion.

#161 · Neuromuscular and Nervous System

Rancho Los Amigos Level I is described how?

Answer

No response, the patient appears in a deep sleep and does not respond to any stimuli.

#162 · Neuromuscular and Nervous System

Rancho Los Amigos Level II is described how?

Answer

Generalized response, the patient reacts inconsistently and non-purposefully to stimuli regardless of the type presented.

#163 · Neuromuscular and Nervous System

Rancho Los Amigos Level III is described how?

Answer

Localized response, the patient reacts specifically but inconsistently to stimuli, such as turning toward a sound.

#164 · Neuromuscular and Nervous System

Rancho Los Amigos Level IV is described how?

Answer

Confused and agitated, the patient is in a heightened state of activity with severely impaired cognition and no goal-directed behavior.

#165 · Neuromuscular and Nervous System

Rancho Los Amigos Level V is described how?

Answer

Confused and inappropriate, non-agitated, the patient can follow simple commands but is highly distractible and lacks goal-directed behavior.

#166 · Neuromuscular and Nervous System

Rancho Los Amigos Level VI is described how?

Answer

Confused and appropriate, the patient shows goal-directed behavior but needs cueing and has impaired recent memory.

#167 · Neuromuscular and Nervous System

Rancho Los Amigos Level VII is described how?

Answer

Automatic and appropriate, the patient performs daily routines automatically but with minimal to no confusion and shallow judgment.

#168 · Neuromuscular and Nervous System

Rancho Los Amigos Level VIII is described how?

Answer

Purposeful and appropriate, the patient is alert and oriented with the ability to recall and integrate past and recent events, though some abstract reasoning deficits may remain.

#169 · Neuromuscular and Nervous System

At which Rancho Los Amigos level should a therapist emphasize a highly structured, low-stimulation environment?

Answer

Level IV, since the patient is confused and agitated and is easily overwhelmed by excess sensory input at this stage.

#170 · Neuromuscular and Nervous System

What is the Glasgow Coma Scale used for and what are its three components?

Answer

It is used to grade level of consciousness after brain injury, scoring eye opening, verbal response, and motor response, with scores ranging from 3 to 15.

#171 · Neuromuscular and Nervous System

A Glasgow Coma Scale score of 8 or less indicates what severity of injury?

Answer

Severe traumatic brain injury, generally consistent with coma.

#172 · Neuromuscular and Nervous System

What is post-traumatic amnesia and why is it clinically important?

Answer

It is the period after brain injury during which the patient cannot form new continuous memories, and its duration correlates with injury severity and prognosis.

#173 · Neuromuscular and Nervous System

What does ASIA Impairment Scale grade A represent?

Answer

Complete injury with no motor or sensory function preserved in the sacral segments S4 to S5.

#174 · Neuromuscular and Nervous System

What does ASIA Impairment Scale grade B represent?

Answer

Sensory incomplete injury, sensory but not motor function is preserved below the neurological level including S4 to S5.

#175 · Neuromuscular and Nervous System

What does ASIA Impairment Scale grade C represent?

Answer

Motor incomplete injury where more than half of key muscles below the neurological level have a muscle grade less than 3.

#176 · Neuromuscular and Nervous System

What does ASIA Impairment Scale grade D represent?

Answer

Motor incomplete injury where at least half of key muscles below the neurological level have a muscle grade of 3 or greater.

#177 · Neuromuscular and Nervous System

What does ASIA Impairment Scale grade E represent?

Answer

Normal sensory and motor function throughout, used when a patient with a prior deficit has recovered fully.

#178 · Neuromuscular and Nervous System

What is the key defining feature that separates a complete from an incomplete spinal cord injury?

Answer

Preservation of sensory or motor function at the sacral segments S4 to S5, including deep anal sensation and voluntary anal contraction.

#179 · Neuromuscular and Nervous System

What functional level of tetraplegia corresponds to preserved C5 function?

Answer

Elbow flexion and shoulder function are present, but the patient lacks wrist extension. This allows independent feeding and grooming with adaptive equipment and a power wheelchair, though the patient requires assistance for bathing, dressing, transfers, and bowel and bladder care.

#180 · Neuromuscular and Nervous System

What functional level of tetraplegia corresponds to preserved C6 function?

Answer

Wrist extensors are present allowing tenodesis grasp, enabling independent transfers with a sliding board and some independence in self-care.

#181 · Neuromuscular and Nervous System

What functional level of tetraplegia corresponds to preserved C7 function?

Answer

Triceps function is present enabling independent transfers, a manual wheelchair for community mobility, and most self-care independence.

#182 · Neuromuscular and Nervous System

What is the expected functional outcome for a complete injury at the thoracic level?

Answer

High thoracic injuries (T1-T6) rely on a wheelchair for community mobility with full upper extremity function and independence in most self-care and transfers. Low thoracic injuries (T10-T12) can achieve household ambulation with bilateral knee-ankle-foot orthoses.

#183 · Neuromuscular and Nervous System

At what level of paraplegia is community ambulation with orthoses generally realistic?

Answer

Injuries at L3 and below, where quadriceps function is preserved, making functional ambulation with ankle-foot orthoses more feasible.

#184 · Neuromuscular and Nervous System

What is autonomic dysreflexia and at what injury level does it typically occur?

Answer

A sudden, dangerous rise in blood pressure with bradycardia, pounding headache, and flushing above the lesion, triggered by a noxious stimulus below the level of injury, typically occurring at T6 and above.

#185 · Neuromuscular and Nervous System

What is the first action a therapist should take when autonomic dysreflexia is suspected?

Answer

Sit the patient upright immediately to lower blood pressure and then search for and remove the noxious stimulus, such as a full bladder or bowel or tight clothing.

#186 · Neuromuscular and Nervous System

What is spinal shock and how long does it typically last?

Answer

A temporary loss of all reflex activity below the level of spinal cord injury, typically resolving within days to a few weeks and marked by the return of the bulbocavernosus reflex.

#187 · Neuromuscular and Nervous System

What is Brown-Sequard syndrome?

Answer

A hemisection of the spinal cord causing ipsilateral loss of motor function and proprioception below the lesion, with contralateral loss of pain and temperature sensation beginning a few levels below.

#188 · Neuromuscular and Nervous System

What is central cord syndrome and its classic presentation?

Answer

Damage centered on the central spinal cord, most common after hyperextension injury in older adults, causing upper extremity weakness that is more pronounced than lower extremity weakness.

#189 · Neuromuscular and Nervous System

What is anterior cord syndrome?

Answer

Damage to the anterior two-thirds of the spinal cord causing loss of motor function, pain, and temperature sensation bilaterally below the lesion, while proprioception and vibration remain intact.

#190 · Neuromuscular and Nervous System

What is the earliest sign of multiple sclerosis relapse most patients report?

Answer

Optic neuritis, presenting as unilateral vision loss or pain with eye movement, is one of the most common initial presentations.

#191 · Neuromuscular and Nervous System

What is Uhthoff's phenomenon in multiple sclerosis?

Answer

Temporary worsening of neurological symptoms with increased body temperature, such as during exercise or a hot shower, that resolves with cooling.

#192 · Neuromuscular and Nervous System

What are the four clinical disease courses of multiple sclerosis?

Answer

Relapsing-remitting, secondary progressive, primary progressive, and progressive relapsing.

#193 · Neuromuscular and Nervous System

What exercise precaution is most important for patients with multiple sclerosis?

Answer

Avoid overheating during exercise, since increased core temperature can transiently worsen symptoms through impaired nerve conduction in demyelinated axons.

#194 · Neuromuscular and Nervous System

What are the four cardinal motor signs of Parkinson's disease?

Answer

Resting tremor, rigidity, bradykinesia, and postural instability.

#195 · Neuromuscular and Nervous System

What type of rigidity is classically seen in Parkinson's disease?

Answer

Cogwheel rigidity, a ratchet-like resistance felt throughout passive range of motion, often combined with lead-pipe rigidity.

#196 · Neuromuscular and Nervous System

What gait pattern is characteristic of Parkinson's disease?

Answer

A festinating gait with short shuffling steps, reduced arm swing, stooped posture, and difficulty initiating movement or turning.

#197 · Neuromuscular and Nervous System

What treatment strategy helps patients with Parkinson's disease overcome freezing of gait?

Answer

External cueing strategies such as visual floor markers, rhythmic auditory cues, or verbal counting to step over rather than relying on internally generated movement.

#198 · Neuromuscular and Nervous System

What is the Hoehn and Yahr scale used for?

Answer

It stages the severity and progression of Parkinson's disease from unilateral involvement only at stage 1 to wheelchair-bound or bedridden at stage 5.

#199 · Neuromuscular and Nervous System

What is Guillain-Barre syndrome and its typical progression pattern?

Answer

An acute autoimmune polyneuropathy causing ascending, symmetric weakness and sensory changes, usually beginning in the distal lower extremities and progressing upward over days to weeks.

#200 · Neuromuscular and Nervous System

What is the primary exercise precaution during the acute phase of Guillain-Barre syndrome?

Answer

Avoid overexertion and high-resistance exercise, since overwork weakness can cause further damage to recovering, denervated muscle fibers.

#201 · Neuromuscular and Nervous System

What distinguishes a peripheral neuropathy sensory pattern from a specific nerve lesion?

Answer

Peripheral neuropathy typically produces a symmetric stocking-glove distribution of sensory loss, while a single nerve lesion follows the distribution of that specific peripheral nerve.

#202 · Neuromuscular and Nervous System

What is the most common presentation of carpal tunnel syndrome?

Answer

Compression of the median nerve at the wrist causing numbness and tingling in the thumb, index, middle, and half of the ring finger, often worse at night.

#203 · Neuromuscular and Nervous System

What muscle weakness pattern indicates a radial nerve palsy?

Answer

Wrist drop from loss of wrist and finger extensors, with sensory loss over the dorsal web space between the thumb and index finger.

#204 · Neuromuscular and Nervous System

What is the sensory distribution of the C6 dermatome?

Answer

The lateral forearm, thumb, and index finger.

#205 · Neuromuscular and Nervous System

What is the sensory distribution of the C7 dermatome?

Answer

The middle finger and central posterior forearm.

#206 · Neuromuscular and Nervous System

What is the sensory distribution of the C8 dermatome?

Answer

The ring and little fingers and the medial (ulnar) hand.

#207 · Neuromuscular and Nervous System

What is the sensory distribution of the L4 dermatome?

Answer

The medial leg and shin down to the medial malleolus.

#208 · Neuromuscular and Nervous System

What is the sensory distribution of the L5 dermatome?

Answer

The lateral leg and dorsum of the foot, including the great toe.

#209 · Neuromuscular and Nervous System

What is the sensory distribution of the S1 dermatome?

Answer

The lateral foot and sole, including the small toe.

#210 · Neuromuscular and Nervous System

Which myotome is tested by resisted knee extension?

Answer

L3, though L2 through L4 all contribute to knee extension strength.

#211 · Neuromuscular and Nervous System

Which myotome is tested by resisted ankle dorsiflexion?

Answer

L4.

#212 · Neuromuscular and Nervous System

Which myotome is tested by resisted great toe extension?

Answer

L5.

#213 · Neuromuscular and Nervous System

Which myotome is tested by resisted ankle plantarflexion or eversion?

Answer

S1.

#214 · Neuromuscular and Nervous System

Which spinal nerve root primarily mediates the biceps deep tendon reflex?

Answer

C5, with contribution from C6.

#215 · Neuromuscular and Nervous System

Which spinal nerve root primarily mediates the brachioradialis deep tendon reflex?

Answer

C6.

#216 · Neuromuscular and Nervous System

Which spinal nerve root primarily mediates the triceps deep tendon reflex?

Answer

C7.

#217 · Neuromuscular and Nervous System

Which spinal nerve root primarily mediates the patellar deep tendon reflex?

Answer

L4, with contribution from L2 through L3.

#218 · Neuromuscular and Nervous System

Which spinal nerve root primarily mediates the Achilles deep tendon reflex?

Answer

S1, with contribution from S2.

#219 · Neuromuscular and Nervous System

What grading is assigned to a deep tendon reflex that is absent even with reinforcement?

Answer

Grade 0, indicating an areflexic response, which typically points to a lower motor neuron lesion.

#220 · Neuromuscular and Nervous System

What deep tendon reflex grade is considered normal?

Answer

Grade 2+, a normal average response.

#221 · Neuromuscular and Nervous System

What does a hyperactive deep tendon reflex with clonus generally indicate?

Answer

An upper motor neuron lesion, since damage to descending inhibitory pathways releases spinal reflex arcs from central control.

#222 · Neuromuscular and Nervous System

What is the Dix-Hallpike test used to diagnose?

Answer

Benign paroxysmal positional vertigo, specifically identifying which posterior or anterior semicircular canal is affected by observing the direction and latency of nystagmus.

#223 · Neuromuscular and Nervous System

How is the Dix-Hallpike test performed?

Answer

The patient is moved quickly from sitting to supine with the head turned 45 degrees to one side and extended 20 degrees over the table edge, and the eyes are observed for nystagmus for at least 30 seconds.

#224 · Neuromuscular and Nervous System

What treatment maneuver is used for posterior canal benign paroxysmal positional vertigo?

Answer

The Epley maneuver, a canalith repositioning technique that moves displaced otoconia out of the posterior semicircular canal and back into the vestibule.

#225 · Neuromuscular and Nervous System

What is the typical nystagmus pattern seen with posterior canal BPPV on the Dix-Hallpike test?

Answer

Upbeating and torsional nystagmus toward the affected ear, with onset delayed a few seconds and fatiguing with repeated testing.

#226 · Neuromuscular and Nervous System

What is the head thrust test used to assess?

Answer

The vestibulo-ocular reflex, identifying a peripheral vestibular hypofunction when a corrective saccade is seen after a rapid head turn toward the affected side.

#227 · Neuromuscular and Nervous System

What is the general goal of vestibular rehabilitation for unilateral hypofunction?

Answer

Promote central compensation through gaze stabilization exercises, habituation exercises, and balance retraining, since the peripheral deficit itself often cannot be reversed.

#228 · Neuromuscular and Nervous System

What is dysdiadochokinesia and what does it indicate?

Answer

Impaired ability to perform rapid alternating movements, indicating cerebellar dysfunction, tested with rapid forearm pronation and supination.

#229 · Neuromuscular and Nervous System

What is dysmetria and how is it tested?

Answer

Impaired ability to judge distance or scale of movement, tested with the finger-to-nose test or heel-to-shin test, indicating cerebellar dysfunction.

#230 · Neuromuscular and Nervous System

What gait pattern is characteristic of cerebellar ataxia?

Answer

A wide-based, staggering gait with irregular step length and lateral veering, often worsened with eyes closed but not as dramatically as with sensory ataxia.

#231 · Neuromuscular and Nervous System

What is the Romberg test used to assess and what does a positive result suggest?

Answer

It assesses reliance on proprioceptive versus visual input for balance, and a positive result, a significant increase in sway or falling with eyes closed, suggests a proprioceptive or vestibular deficit rather than cerebellar dysfunction.

#232 · Neuromuscular and Nervous System

What is intention tremor and which condition is it associated with?

Answer

A tremor that worsens as a limb approaches its target during a voluntary movement, associated with cerebellar dysfunction, as opposed to the resting tremor of Parkinson's disease.

#233 · Neuromuscular and Nervous System

What does the Berg Balance Scale assess and what score suggests high fall risk?

Answer

It assesses functional balance across 14 tasks scored 0 to 4, with a total score below 45 out of 56 generally indicating an increased risk of falling.

#234 · Neuromuscular and Nervous System

What does the Timed Up and Go test measure and what cutoff suggests fall risk?

Answer

It measures the time to rise from a chair, walk 3 meters, turn, return, and sit down, with times greater than 12 seconds generally associated with increased fall risk in older adults.

#235 · Neuromuscular and Nervous System

What does the Functional Reach Test measure and what cutoff suggests fall risk?

Answer

It measures the maximum distance a person can reach forward beyond arm's length while maintaining a fixed base of support, with a reach of less than 6 inches indicating high fall risk.

#236 · Neuromuscular and Nervous System

What is the Dynamic Gait Index used to assess?

Answer

It assesses a person's ability to modify gait in response to changing task demands, such as walking with head turns, over obstacles, or with pivot turns, across 8 items.

#237 · Neuromuscular and Nervous System

What is the four-square step test used to assess?

Answer

Dynamic balance and stepping ability by having the patient step over four canes or sticks arranged in a cross pattern as quickly and safely as possible in all directions.

#238 · Neuromuscular and Nervous System

What is the core principle of neurodevelopmental treatment approach?

Answer

Facilitating normal movement patterns and inhibiting abnormal tone or reflexes through handling and key points of control, guiding the patient toward more typical postural and movement responses.

#239 · Neuromuscular and Nervous System

What is a key point of control in neurodevelopmental treatment?

Answer

A proximal body region, such as the shoulder, pelvis, or trunk, that a therapist uses to guide and influence movement quality throughout the rest of the body.

#240 · Neuromuscular and Nervous System

What is weight bearing used for in neurodevelopmental treatment approaches?

Answer

To normalize muscle tone, provide proprioceptive input, and facilitate postural stability and co-contraction around a joint or limb.

#241 · Neuromuscular and Nervous System

What is the tonic labyrinthine reflex and how does it present?

Answer

A primitive reflex in which prone positioning increases flexor tone and supine positioning increases extensor tone throughout the body.

#242 · Neuromuscular and Nervous System

What is the asymmetrical tonic neck reflex and how does it present?

Answer

Turning the head to one side causes extension of the limbs on the face side and flexion of the limbs on the skull side, resembling a fencing posture.

#243 · Neuromuscular and Nervous System

What is the symmetrical tonic neck reflex and how does it present?

Answer

Neck flexion causes arm flexion and leg extension, while neck extension causes arm extension and leg flexion.

#244 · Neuromuscular and Nervous System

Why is persistence of primitive reflexes beyond infancy clinically significant?

Answer

It suggests central nervous system immaturity or damage and can interfere with the development of normal voluntary movement and functional postural control.

#245 · Neuromuscular and Nervous System

What is decorticate posturing and what does it indicate?

Answer

Upper extremity flexion with lower extremity extension, indicating a lesion above the level of the red nucleus in the midbrain.

#246 · Neuromuscular and Nervous System

What is decerebrate posturing and what does it indicate?

Answer

Rigid extension of both the upper and lower extremities, indicating a more caudal brainstem lesion at or below the level of the red nucleus, generally a worse prognostic sign than decorticate posturing.

#247 · Neuromuscular and Nervous System

What are early signs of increased intracranial pressure that a therapist should monitor for?

Answer

Headache, vomiting, decreased level of consciousness, and Cushing's triad of hypertension, bradycardia, and irregular respiration.

#248 · Neuromuscular and Nervous System

What is heterotopic ossification and in which populations is it common?

Answer

Abnormal bone formation in soft tissue around joints, commonly seen after traumatic brain injury and spinal cord injury, often presenting as decreased range of motion with local swelling and warmth.

#249 · Neuromuscular and Nervous System

What is the difference between flaccidity and spasticity in terms of muscle tone?

Answer

Flaccidity is complete absence of muscle tone typically from a lower motor neuron or acute upper motor neuron lesion, while spasticity is velocity-dependent increased resistance to passive stretch from an upper motor neuron lesion.

#250 · Neuromuscular and Nervous System

What is clonus and how is it clinically tested?

Answer

Rhythmic, involuntary muscle contractions in response to a sudden sustained stretch, commonly tested at the ankle by quickly dorsiflexing the foot and observing rhythmic beats.

#251 · Neuromuscular and Nervous System

What is the Modified Ashworth Scale used to measure?

Answer

The degree of resistance to passive stretch as a clinical measure of spasticity, graded from 0, no increase in tone, to 4, rigid in flexion or extension.

#252 · Neuromuscular and Nervous System

What distinguishes upper motor neuron from lower motor neuron lesion presentations?

Answer

Upper motor neuron lesions cause spasticity, hyperreflexia, and no significant atrophy, while lower motor neuron lesions cause flaccidity, hyporeflexia or areflexia, and significant muscle atrophy with fasciculations.

#253 · Neuromuscular and Nervous System

What is myasthenia gravis and its characteristic pattern of weakness?

Answer

An autoimmune disorder affecting the neuromuscular junction that causes fluctuating muscle weakness that worsens with activity and improves with rest, often affecting ocular and facial muscles first.

#254 · Neuromuscular and Nervous System

What is amyotrophic lateral sclerosis and what motor neuron pattern does it affect?

Answer

A progressive neurodegenerative disease affecting both upper and lower motor neurons, presenting with a combination of spasticity and flaccidity, fasciculations, and progressive muscle wasting, without sensory loss.

#255 · Neuromuscular and Nervous System

What is the primary treatment consideration for exercise prescription in amyotrophic lateral sclerosis?

Answer

Use moderate, submaximal exercise intensity and avoid overwork fatigue, since excessive exertion in already-compromised motor units can accelerate weakness.

#256 · Neuromuscular and Nervous System

What is diabetic peripheral neuropathy's typical clinical presentation?

Answer

A symmetric, distal, stocking-glove sensory loss beginning in the feet, with decreased proprioception and vibration sense increasing fall risk and skin breakdown risk.

#257 · Neuromuscular and Nervous System

What is the pull test used for in Parkinson's disease assessment?

Answer

It assesses postural instability by applying a quick backward pull on the shoulders while the examiner stands ready to catch the patient, with more than two retropulsive steps or a fall considered abnormal.

#258 · Neuromuscular and Nervous System

What is the primary difference between an upper motor neuron bladder and a lower motor neuron bladder after spinal cord injury?

Answer

An upper motor neuron bladder is spastic and reflexively empties involuntarily, while a lower motor neuron bladder is flaccid, retains urine, and requires manual expression or catheterization.

#259 · Neuromuscular and Nervous System

What is the significance of a positive Babinski sign?

Answer

Dorsiflexion of the great toe with fanning of the other toes upon plantar stimulation indicates an upper motor neuron lesion in anyone beyond infancy.

#260 · Neuromuscular and Nervous System

What is apraxia and how does it differ from paresis?

Answer

Apraxia is the inability to perform a learned purposeful motor task despite intact strength, sensation, and comprehension, distinguishing it from paresis which is true muscle weakness.

#261 • Cardiovascular and Pulmonary Systems

What are the goals of Phase I cardiac rehabilitation?

Answer

Phase I is the inpatient acute phase after MI, CABG, or other cardiac event. Goals include early mobilization, patient education, prevention of deconditioning, and safe progression of low-level activity (typically 1 to 2 METs) before discharge.

#262 • Cardiovascular and Pulmonary Systems

What defines Phase II cardiac rehabilitation?

Answer

Phase II is an outpatient, monitored program that typically begins within a few weeks of discharge and lasts 8 to 12 weeks. It involves ECG-monitored exercise, risk factor modification, and gradual progression of exercise intensity.

#263 · Cardiovascular and Pulmonary Systems

What characterizes Phase III cardiac rehabilitation?

Answer

Phase III is a supervised but typically non-ECG-monitored community-based exercise program that continues to build endurance and independence, usually lasting several months after Phase II completion.

#264 · Cardiovascular and Pulmonary Systems

What is Phase IV cardiac rehabilitation?

Answer

Phase IV is a lifelong, unsupervised maintenance phase focused on independent exercise, ongoing risk factor management, and long-term lifestyle change to sustain cardiovascular health.

#265 • Cardiovascular and Pulmonary Systems

What is the numeric range of the original Borg RPE scale?

Answer

The original Borg scale ranges from 6 to 20, with the numbers roughly correlating to heart rate divided by 10 (for example, a rating of 13 corresponds to about a heart rate of 130 bpm in an average adult).

#266 • Cardiovascular and Pulmonary Systems

What RPE range on the 6 to 20 Borg scale is generally recommended for moderate-intensity cardiac rehab exercise?

Answer

A rating of 12 to 13 (somewhat hard) on the 6 to 20 scale is generally the target for moderate-intensity aerobic exercise in cardiac rehabilitation.

#267 · Cardiovascular and Pulmonary Systems

What is the range of the modified (category-ratio) Borg CR10 scale?

Answer

The modified Borg CR10 scale ranges from 0 (nothing at all) to 10 (maximal exertion), and a rating of 3 to 4 is often used as a target for moderate exercise intensity.

#268 · Cardiovascular and Pulmonary Systems

What does the F in the FITT principle stand for, and what does it mean?

Answer

F stands for Frequency, referring to how often exercise is performed, typically expressed as sessions per week (for example, 3 to 5 days per week for aerobic conditioning).

#269 · Cardiovascular and Pulmonary Systems

What does the I in the FITT principle stand for?

Answer

I stands for Intensity, the level of exertion during exercise, which can be prescribed using percentage of heart rate reserve, RPE, MET level, or percentage of VO₂ max.

#270 · Cardiovascular and Pulmonary Systems

What does the first T in the FITT principle stand for?

Answer

The first T stands for Time, meaning the duration of each exercise session, commonly 20 to 60 minutes for cardiovascular conditioning.

#271 • Cardiovascular and Pulmonary Systems

What does the second T in the FITT principle stand for?

Answer

The second T stands for Type, meaning the mode of exercise such as walking, cycling, swimming, or resistance training, chosen based on patient goals and limitations.

#272 • Cardiovascular and Pulmonary Systems

How is target heart rate calculated using the Karvonen (heart rate reserve) formula?

Answer

Target HR equals (heart rate reserve times desired intensity percentage) plus resting heart rate, where heart rate reserve equals maximum heart rate minus resting heart rate.

#273 · Cardiovascular and Pulmonary Systems

How is age-predicted maximum heart rate estimated?

Answer

Age-predicted maximum heart rate is commonly estimated as 220 minus the patient's age in years, though this is a general estimate and can vary considerably between individuals.

#274 · Cardiovascular and Pulmonary Systems

What is one MET defined as?

Answer

One MET (metabolic equivalent of task) is defined as the resting oxygen consumption of a seated adult, equal to approximately 3.5 mL of oxygen per kilogram of body weight per minute.

#275 • Cardiovascular and Pulmonary Systems

Approximately how many METs does slow walking (about 2 mph) require?

Answer

Slow, level walking at about 2 mph requires approximately 2 to 2.5 METs, generally considered a low-intensity activity appropriate for early Phase I or II cardiac rehab.

#276 • Cardiovascular and Pulmonary Systems

Approximately how many METs does light housework such as dusting or dishwashing require?

Answer

Light housework typically requires approximately 2 to 3 METs, similar in demand to slow walking.

#277 · Cardiovascular and Pulmonary Systems

Approximately how many METs does climbing stairs require?

Answer

Climbing stairs at a normal pace typically requires approximately 4 to 6 METs, rising toward 8 METs when climbing rapidly or carrying a load.

#278 · Cardiovascular and Pulmonary Systems

Approximately how many METs does sexual activity typically require?

Answer

Sexual activity with a familiar partner typically requires approximately 2.5 to 5 METs, a common benchmark used when counseling patients recovering from a cardiac event.

#279 • Cardiovascular and Pulmonary Systems

What MET level is often used as the criterion for safe hospital discharge after MI?

Answer

Tolerating an activity level of approximately 3.5 to 5 METs without abnormal signs or symptoms is often used as a benchmark for readiness for hospital discharge after an uncomplicated MI.

#280 • Cardiovascular and Pulmonary Systems

What blood pressure response to exercise is considered abnormal and requires terminating a test or session?

Answer

A drop in systolic blood pressure of 10 mmHg or more from baseline that occurs with an increase in workload, especially when accompanied by other signs of ischemia, is an abnormal response and an indication to stop exercise.

#281 • Cardiovascular and Pulmonary Systems

What resting or exercise blood pressure values generally indicate exercise should not begin or should be terminated?

Answer

A systolic blood pressure greater than 250 mmHg or diastolic greater than 115 mmHg, or a resting systolic below 90 mmHg with symptoms, are generally indications to withhold or stop exercise.

#282 • Cardiovascular and Pulmonary Systems

What heart rate response is considered abnormal during exercise testing?

Answer

A failure of heart rate to increase with increasing workload (chronotropic incompetence), or an excessive rise in heart rate disproportionate to workload, are both considered abnormal responses warranting further evaluation.

#283 • Cardiovascular and Pulmonary Systems

What oxygen saturation level is commonly used as a criterion to terminate an activity or exercise session?

Answer

An SpO₂ that falls below 88 to 90 percent is a common criterion to stop activity and reassess the patient, since this indicates significant desaturation.

#284 • Cardiovascular and Pulmonary Systems

What signs and symptoms should prompt immediate termination of an exercise session regardless of vital signs?

Answer

Onset of angina, significant dyspnea, dizziness, pallor, cyanosis, ataxia, confusion, or patient request to stop should all prompt immediate cessation of exercise.

#285 • Cardiovascular and Pulmonary Systems

What are typical post-MI or post-CABG activity precautions in the acute phase?

Answer

Patients are typically advised to avoid heavy lifting, Valsalva maneuvers, and high-intensity upper extremity activity, and sternal precautions after CABG restrict pushing, pulling, or lifting more than about 5 to 10 pounds for approximately 6 to 8 weeks.

#286 • Cardiovascular and Pulmonary Systems

What are sternal precautions and why are they used?

Answer

Sternal precautions limit shoulder flexion and abduction beyond 90 degrees, bilateral overhead reaching, and pushing or pulling with the arms, to protect the healing sternotomy incision and prevent sternal instability after CABG surgery.

#287 · Cardiovascular and Pulmonary Systems

What is claudication?

Answer

Claudication is reproducible ischemic muscle pain, most commonly in the calf, that occurs with a consistent level of activity such as walking and is relieved by rest, caused by inadequate arterial blood flow (peripheral vascular disease).

#288 · Cardiovascular and Pulmonary Systems

What exercise recommendation is typically given for patients with intermittent claudication from PVD?

Answer

Patients are typically instructed to walk to a moderate level of claudication pain, rest until it subsides, then resume walking, repeating this interval pattern to promote collateral circulation and improve walking tolerance over time.

#289 · Cardiovascular and Pulmonary Systems

What is the ankle-brachial index (ABI) used for?

Answer

The ABI is the ratio of ankle systolic blood pressure to brachial systolic blood pressure, used to screen for and grade the severity of peripheral arterial disease; a normal ABI is approximately 1.0 to 1.4.

#290 · Cardiovascular and Pulmonary Systems

What ABI value is generally considered indicative of significant peripheral arterial disease?

Answer

An ABI below 0.9 is generally considered indicative of peripheral arterial disease, with values below 0.4 suggesting severe disease and possible critical limb ischemia.

#291 • Cardiovascular and Pulmonary Systems

What are the classic signs of a deep vein thrombosis (DVT)?

Answer

Classic signs of DVT include unilateral calf or leg swelling, warmth, redness, and tenderness, sometimes with a palpable cord; Homans sign is unreliable and no longer recommended for clinical use.

#292 • Cardiovascular and Pulmonary Systems

What should a therapist do if a DVT is suspected during a treatment session?

Answer

If DVT is suspected, the therapist should stop exercise, avoid massaging or aggressively mobilizing the limb, and notify the physician immediately due to the risk of pulmonary embolism from a dislodged clot.

#293 · Cardiovascular and Pulmonary Systems

What are common signs and symptoms of a pulmonary embolism?

Answer

Common signs and symptoms of pulmonary embolism include sudden onset dyspnea, pleuritic chest pain, tachycardia, tachypnea, hemoptysis, and anxiety or a sense of impending doom, and it represents a medical emergency.

#294 · Cardiovascular and Pulmonary Systems

What is the hallmark pulmonary function pattern of COPD?

Answer

COPD demonstrates an obstructive pattern with a reduced FEV1 to FVC ratio, increased total lung capacity and residual volume from air trapping, and reduced expiratory flow rates.

#295 • Cardiovascular and Pulmonary Systems

What is the hallmark pulmonary function pattern of restrictive lung disease?

Answer

Restrictive lung disease shows a reduced total lung capacity and reduced forced vital capacity, but the FEV1 to FVC ratio is normal or even increased because both values fall proportionally.

#296 • Cardiovascular and Pulmonary Systems

What is pursed-lip breathing used for in COPD management?

Answer

Pursed-lip breathing slows expiratory flow and creates back-pressure that helps keep airways open longer during exhalation, reducing air trapping, dyspnea, and the work of breathing in patients with COPD.

#297 · Cardiovascular and Pulmonary Systems

What is diaphragmatic (belly) breathing used to teach patients with pulmonary disease?

Answer

Diaphragmatic breathing trains patients to use the diaphragm rather than accessory muscles for inspiration, improving ventilation efficiency and reducing the work of breathing, particularly useful in COPD and post-surgical patients.

#298 · Cardiovascular and Pulmonary Systems

What position is commonly used to relieve dyspnea in patients with COPD?

Answer

The tripod position, sitting leaning forward with arms braced on the knees or a table, is commonly used because it stabilizes the shoulder girdle and allows accessory muscles to assist more effectively with breathing.

#299 · Cardiovascular and Pulmonary Systems

What are the goals of pulmonary rehabilitation?

Answer

Pulmonary rehabilitation aims to improve exercise tolerance, reduce dyspnea, improve breathing efficiency, provide education on disease self-management, and enhance overall quality of life in patients with chronic pulmonary disease.

#300 · Cardiovascular and Pulmonary Systems

What is postural drainage in chest physical therapy?

Answer

Postural drainage uses specific gravity-assisted body positions to promote the movement of secretions from particular lung segments toward the central airways for easier clearance by coughing or suctioning.

#301 • Cardiovascular and Pulmonary Systems

What is percussion in chest physical therapy and what is its purpose?

Answer

Percussion involves rhythmic clapping with cupped hands over the chest wall to mechanically loosen retained secretions, typically performed in conjunction with postural drainage positioning.

#302 • Cardiovascular and Pulmonary Systems

What is vibration in chest physical therapy and how does it differ from percussion?

Answer

Vibration is a fine oscillatory pressure applied with flattened hands over the chest wall during exhalation to help move loosened secretions toward larger airways; unlike percussion, it is applied only on exhalation.

#303 • Cardiovascular and Pulmonary Systems

What is huffing (forced expiratory technique) used for in airway clearance?

Answer

Huffing involves a forced but open-glottis exhalation used to move secretions toward the central airways without triggering the airway collapse or fatigue that repeated forceful coughing can cause.

#304 • Cardiovascular and Pulmonary Systems

What contraindications should be checked before performing postural drainage with head-down (Trendelenburg) positions?

Answer

Head-down postural drainage positions are contraindicated or used cautiously in patients with uncontrolled hypertension, increased intracranial pressure, recent eye or abdominal surgery, severe dyspnea, or hemoptysis.

#305 · Cardiovascular and Pulmonary Systems

What is the incentive spirometer used for and how is it used?

Answer

The incentive spirometer is a device that encourages slow, sustained maximal inspiration to prevent or reverse atelectasis, commonly used postoperatively; the patient inhales slowly to raise the indicator ball or piston and holds the breath briefly before exhaling.

#306 · Cardiovascular and Pulmonary Systems

What is a normal P wave on an ECG represent?

Answer

The P wave represents atrial depolarization, and a normal P wave should be upright, rounded, and precede each QRS complex in a regular sinus rhythm.

#307 · Cardiovascular and Pulmonary Systems

What does the QRS complex on an ECG represent?

Answer

The QRS complex represents ventricular depolarization, with a normal duration of less than 0.12 seconds; a widened QRS suggests a conduction delay such as a bundle branch block or ventricular origin of the beat.

#308 · Cardiovascular and Pulmonary Systems

What does the T wave on an ECG represent?

Answer

The T wave represents ventricular repolarization, and inverted or peaked T waves can indicate myocardial ischemia or electrolyte disturbances such as hyperkalemia.

#309 • Cardiovascular and Pulmonary Systems

What ECG finding is characteristic of atrial fibrillation?

Answer

Atrial fibrillation is characterized by an irregularly irregular rhythm with no discernible, consistent P waves, replaced instead by a chaotic fibrillatory baseline.

#310 • Cardiovascular and Pulmonary Systems

What defines sinus tachycardia on an ECG?

Answer

Sinus tachycardia shows a normal P wave before each QRS complex with a regular rhythm, but the rate exceeds 100 beats per minute, often a normal physiologic response to exercise, fever, or stress.

#311 · Cardiovascular and Pulmonary Systems

What is ventricular tachycardia and why is it clinically significant?

Answer

Ventricular tachycardia is a rapid rhythm of three or more consecutive wide QRS complexes originating in the ventricles, which is clinically significant because it can rapidly deteriorate into ventricular fibrillation and cardiac arrest, requiring immediate medical attention.

#312 · Cardiovascular and Pulmonary Systems

What does ST segment elevation on an ECG typically indicate?

Answer

ST segment elevation typically indicates acute myocardial injury, most classically an ST-elevation myocardial infarction (STEMI), and requires immediate medical evaluation.

#313 • Cardiovascular and Pulmonary Systems

What does ST segment depression typically indicate, particularly during exercise testing?

Answer

ST segment depression, especially horizontal or downsloping depression of 1 mm or more during exercise, typically indicates myocardial ischemia and is a criterion to terminate an exercise test.

#314 • Cardiovascular and Pulmonary Systems

What is a premature ventricular contraction (PVC) and when does it become concerning?

Answer

A PVC is an early, wide QRS complex originating from an ectopic ventricular focus without a preceding P wave; PVCs become concerning when they occur frequently, in couplets or triplets, or increase with exercise, since this may signal ventricular irritability.

#315 • Cardiovascular and Pulmonary Systems

What activity should be avoided immediately after cardiac catheterization via the femoral approach?

Answer

Hip flexion, sitting upright, and ambulation should be avoided for a specified period after femoral catheterization to prevent bleeding or hematoma formation at the puncture site, per physician protocol.

#316 • Cardiovascular and Pulmonary Systems

What is the general rule of thumb for progressing MET level demands during early cardiac rehabilitation?

Answer

A common general guideline is not to exceed the MET level achieved during symptom-limited exercise testing, and to progress gradually while monitoring for signs of overexertion such as excessive fatigue lasting into the next day.

#317 · Cardiovascular and Pulmonary Systems

What is angina pectoris?

Answer

Angina pectoris is chest discomfort or pain caused by reversible myocardial ischemia, typically brought on by exertion or stress and relieved by rest or nitroglycerin.

#318 · Cardiovascular and Pulmonary Systems

How does stable angina differ from unstable angina?

Answer

Stable angina occurs predictably with a consistent level of exertion and resolves with rest or nitroglycerin, while unstable angina occurs at rest, is new in onset, or is increasing in frequency or severity, and represents an acute coronary syndrome requiring urgent medical attention.

#319 • Cardiovascular and Pulmonary Systems

What should a therapist do if a patient reports new or worsening chest pain during exercise?

Answer

The therapist should stop the activity immediately, have the patient rest, assess vital signs, ask about pain characteristics, and notify medical staff promptly, since this could indicate myocardial ischemia.

#320 • Cardiovascular and Pulmonary Systems

What are the FEV1/FVC criteria typically used to grade the severity of obstructive lung disease?

Answer

An FEV1/FVC ratio below 70 percent (0.70) defines obstruction, with severity further graded by the percent-predicted FEV1 value itself, where lower percentages indicate more severe obstruction.

#321 · Cardiovascular and Pulmonary Systems

What is orthopnea and what condition is it commonly associated with?

Answer

Orthopnea is dyspnea that occurs when lying flat and improves with sitting or standing upright, commonly associated with congestive heart failure due to redistribution of fluid to the lungs when supine.

#322 · Cardiovascular and Pulmonary Systems

What is paroxysmal nocturnal dyspnea (PND)?

Answer

PND is sudden, severe shortness of breath that awakens a patient from sleep, typically relieved by sitting up, and is a classic symptom of left-sided heart failure.

#323 • Cardiovascular and Pulmonary Systems

What is a normal resting adult respiratory rate range?

Answer

A normal resting adult respiratory rate is approximately 12 to 20 breaths per minute; tachypnea refers to rates above this range and bradypnea to rates below it.

#324 • Cardiovascular and Pulmonary Systems

What weight gain finding should prompt concern for fluid retention in a patient with heart failure?

Answer

A weight gain of 2 to 3 pounds in a single day, or 5 pounds in a week, should prompt concern for fluid retention and worsening heart failure, warranting physician notification.

#325 • Cardiovascular and Pulmonary Systems

What is the significance of pitting edema in a patient with cardiac disease?

Answer

Pitting edema, particularly in the lower extremities, suggests fluid overload and can indicate right-sided or biventricular heart failure, and its severity is often graded on a scale from 1+ to 4+.

#326 • Cardiovascular and Pulmonary Systems

What is the general precaution regarding resistance training intensity early in cardiac rehabilitation?

Answer

Resistance training in early cardiac rehab typically uses low resistance with higher repetitions, avoiding heavy lifting and Valsalva maneuvers, and is often delayed several weeks after an acute cardiac event per physician clearance.

#327 • Cardiovascular and Pulmonary Systems

What does a decrease in oxygen saturation with exertion in a patient with restrictive or obstructive lung disease typically indicate?

Answer

A drop in SpO₂ with exertion typically indicates a ventilation-perfusion mismatch or impaired gas exchange, and supplemental oxygen titration may be needed to maintain safe saturation during activity.

#328 • Cardiovascular and Pulmonary Systems

What is the general recommendation for exercising a patient who is on supplemental oxygen?

Answer

The patient should continue using their prescribed supplemental oxygen during exercise, with flow rate adjusted per physician order as needed to maintain SpO₂ at or above the target threshold, commonly 90 percent or higher.

#329 • Cardiovascular and Pulmonary Systems

What is the clinical significance of a rate-pressure product (double product) in cardiac rehab monitoring?

Answer

The rate-pressure product, calculated as heart rate multiplied by systolic blood pressure, estimates myocardial oxygen demand, and tracking it helps identify the workload at which a patient develops ischemic symptoms so future sessions can be dosed safely below that threshold.

#330 • Cardiovascular and Pulmonary Systems

What is the general precaution regarding Valsalva maneuver during exercise in cardiac patients?

Answer

Patients should be instructed to avoid breath-holding and straining (Valsalva maneuver) during exertion, such as while lifting, because it causes a rapid rise in intrathoracic pressure followed by a reflexive drop in blood pressure that can lead to dizziness or cardiac strain, and instead should be taught to exhale during the effort phase.

#331 · Lymphatic Metabolic Endocrine GI and GU Systems

What defines Stage 0 (latent) lymphedema in the International Society of Lymphology staging system?

Answer

Subclinical impairment of lymphatic transport with no visible or measurable swelling, though the patient may report heaviness or fullness in the limb. Swelling can take months to years to appear after this stage.

#332 · Lymphatic Metabolic Endocrine GI and GU Systems

What distinguishes Stage I lymphedema from Stage 0?

Answer

Stage I is early accumulation of protein-rich fluid that is relatively high in fluid content and pits on pressure. It reduces with limb elevation, unlike later stages.

#333 · Lymphatic Metabolic Endocrine GI and GU Systems

What are the hallmark features of Stage II lymphedema?

Answer

Limb elevation alone rarely reduces the swelling, and pitting is present initially but tissue fibrosis develops over time so the limb may become non-pitting later in this stage.

#334 · Lymphatic Metabolic Endocrine GI and GU Systems

What characterizes Stage III lymphedema (lymphostatic elephantiasis)?

Answer

Tissue is non-pitting with significant fibrosis and skin changes such as thickening, hyperkeratosis, papillomas, and increased limb volume. It represents the most advanced and least reversible stage.

#335 · Lymphatic Metabolic Endocrine GI and GU Systems

What are the four components of Complete Decongestive Therapy (CDT)?

Answer

Manual lymphatic drainage, compression bandaging or garments, therapeutic exercise, and meticulous skin and nail care.

#336 · Lymphatic Metabolic Endocrine GI and GU Systems

What are the two phases of Complete Decongestive Therapy?

Answer

Phase 1 is the intensive decongestive phase using daily manual lymphatic drainage and short-stretch bandaging to reduce volume. Phase 2 is the maintenance phase using compression garments and self-management to sustain the reduction.

#337 · Lymphatic Metabolic Endocrine GI and GU Systems

Why are short-stretch (low-stretch) bandages preferred over long-stretch elastic bandages in lymphedema management?

Answer

Short-stretch bandages provide high working pressure during muscle contraction to promote lymphatic and venous return, but low resting pressure at rest, which is safer and more effective than the sustained high resting pressure of long-stretch elastic wraps.

#338 · Lymphatic Metabolic Endocrine GI and GU Systems

What direction does manual lymphatic drainage massage move fluid in a limb with lymphedema?

Answer

MLD strokes move fluid distal to proximal, following the normal direction of lymphatic flow toward the trunk and functioning nodes. Treatment sequencing, however, begins proximally, clearing the trunk quadrants and regional nodes first to create a path, before working distally down the limb.

#339 · Lymphatic Metabolic Endocrine GI and GU Systems

What is a key contraindication to manual lymphatic drainage and compression therapy?

Answer

Active untreated infection or cellulitis in the limb, acute deep vein thrombosis, and uncontrolled congestive heart failure, since compression can overload central circulation.

#340 · Lymphatic Metabolic Endocrine GI and GU Systems

What skin finding suggests cellulitis in a lymphedematous limb and requires holding compression therapy?

Answer

Acute onset of warmth, redness, tenderness, and often fever or chills, since compression during active infection can worsen the spread of infection.

#341 · Lymphatic Metabolic Endocrine GI and GU Systems

Why is a diabetic patient at risk of hypoglycemia during exercise if insulin was injected into a working limb?

Answer

Exercise increases blood flow to the injected area, accelerating insulin absorption and action, which can cause a rapid drop in blood glucose during or shortly after activity.

#342 · Lymphatic Metabolic Endocrine GI and GU Systems

What blood glucose level should prompt a physical therapist to postpone exercise in a person with type 1 diabetes?

Answer

Blood glucose below 100 mg/dL should prompt a carbohydrate snack before exercising, and blood glucose above 250 mg/dL with ketones present should prompt postponing exercise due to risk of diabetic ketoacidosis.

#343 · Lymphatic Metabolic Endocrine GI and GU Systems

What are the classic signs of hypoglycemia during an exercise session?

Answer

Shakiness, sweating, confusion, irritability, tachycardia, and pallor. Severe hypoglycemia can progress to loss of consciousness if untreated.

#344 · Lymphatic Metabolic Endocrine GI and GU Systems

What is the recommended immediate treatment for symptomatic hypoglycemia during a therapy session?

Answer

Give 15 to 20 grams of fast-acting carbohydrate such as glucose tablets or juice, recheck blood glucose in 15 minutes, and repeat if still low before resuming activity.

#345 · Lymphatic Metabolic Endocrine GI and GU Systems

Why should patients with diabetic peripheral neuropathy avoid high-impact weight-bearing exercise on insensate feet?

Answer

Loss of protective sensation increases the risk of unnoticed trauma, blisters, and ulceration, so low-impact activities such as cycling or swimming are safer alternatives.

#346 · Lymphatic Metabolic Endocrine GI and GU Systems

What daily foot care instruction should be given to a patient with diabetes before starting an exercise program?

Answer

Inspect the feet daily for redness, blisters, or breaks in skin, wear well-fitting shoes with clean dry socks, and report any nonhealing wound to a physician promptly.

#347 · Lymphatic Metabolic Endocrine GI and GU Systems

How does proliferative diabetic retinopathy affect exercise prescription?

Answer

High-intensity or Valsalva-type activities and activities involving jarring or head-down positions should be avoided because they can increase intraocular pressure and risk vitreous hemorrhage or retinal detachment.

#348 · Lymphatic Metabolic Endocrine GI and GU Systems

What cardiovascular precaution applies to exercise testing in patients with long-standing diabetes?

Answer

Autonomic neuropathy may blunt the normal heart rate response to exercise, so use rating of perceived exertion rather than target heart rate alone to gauge intensity.

#349 · Lymphatic Metabolic Endocrine GI and GU Systems

What are the classic signs and symptoms of hyperthyroidism relevant to exercise tolerance?

Answer

Tachycardia, heat intolerance, weight loss, tremor, anxiety, and exercise intolerance from an elevated resting metabolic rate. Exercise capacity may be reduced despite a hyperdynamic circulation.

#350 · Lymphatic Metabolic Endocrine GI and GU Systems

What are the classic signs and symptoms of hypothyroidism relevant to exercise tolerance?

Answer

Fatigue, cold intolerance, weight gain, muscle weakness and stiffness, bradycardia, and reduced exercise capacity due to a slowed metabolic rate.

#351 · Lymphatic Metabolic Endocrine GI and GU Systems

Why might a patient with untreated hypothyroidism report myalgias and elevated creatine kinase?

Answer

Hypothyroid myopathy can cause muscle weakness, cramping, and elevated creatine kinase from impaired muscle metabolism, which can be mistaken for exercise-induced muscle injury.

#352 · Lymphatic Metabolic Endocrine GI and GU Systems

What is thyroid storm and why is it an exercise contraindication?

Answer

Thyroid storm is a life-threatening exacerbation of hyperthyroidism with severe tachycardia, fever, and altered mental status. Any physical stress including exercise can precipitate or worsen it, so exercise is contraindicated until medically stabilized.

#353 · Lymphatic Metabolic Endocrine GI and GU Systems

What monitoring consideration applies to a patient on levothyroxine who reports new chest pain during exercise?

Answer

Overreplacement of thyroid hormone can cause tachyarrhythmias and angina, so new cardiac symptoms should prompt stopping exercise and physician notification rather than assuming deconditioning.

#354 · Lymphatic Metabolic Endocrine GI and GU Systems

What type of exercise is most effective for increasing bone mineral density in osteoporosis?

Answer

Weight-bearing and resistance exercise that loads bone above habitual stress, such as walking, stair climbing, and progressive resistance training, stimulates osteogenesis more than non-weight-bearing activity.

#355 · Lymphatic Metabolic Endocrine GI and GU Systems

What spinal movements should be avoided in a patient with osteoporosis due to vertebral compression fracture risk?

Answer

Forward flexion combined with loading, such as toe touches or trunk flexion sit-ups, should be avoided since flexion increases anterior vertebral body compressive forces.

#356 · Lymphatic Metabolic Endocrine GI and GU Systems

Why is trunk extension exercise emphasized over trunk flexion exercise in osteoporosis programs?

Answer

Extension exercises strengthen the paraspinal muscles that support upright posture and reduce compressive load on the anterior vertebral bodies, lowering the risk of compression fractures.

#357 · Lymphatic Metabolic Endocrine GI and GU Systems

What activities are generally contraindicated for a patient with severe osteoporosis and multiple vertebral fractures?

Answer

High-impact activities, spinal rotation with loading, and any activity with high fall risk should be avoided, since fracture risk is significantly elevated with minimal trauma.

#358 · Lymphatic Metabolic Endocrine GI and GU Systems

What is the FRAX tool used for in osteoporosis management?

Answer

It estimates a patient's 10-year probability of major osteoporotic fracture and hip fracture based on clinical risk factors and bone mineral density, guiding treatment and exercise decisions.

#359 · Lymphatic Metabolic Endocrine GI and GU Systems

What T-score on DEXA scan defines osteoporosis versus osteopenia?

Answer

A T-score of -2.5 or lower defines osteoporosis, while a T-score between -1.0 and -2.5 defines osteopenia (low bone mass), compared to a healthy young adult reference.

#360 · Lymphatic Metabolic Endocrine GI and GU Systems

What activity precaution follows abdominal surgery in the first several weeks postoperatively?

Answer

Avoid heavy lifting and straining that increases intra-abdominal pressure, since this can disrupt fascial healing and increase risk of incisional hernia.

#361 · Lymphatic Metabolic Endocrine GI and GU Systems

What is the recommended lifting restriction typically given after abdominal surgery?

Answer

Commonly no lifting greater than 10 pounds for the first several weeks, though the exact limit depends on surgeon protocol and type of procedure.

#362 · Lymphatic Metabolic Endocrine GI and GU Systems

How should a patient with a colostomy be positioned or braced during therapeutic exercise involving trunk movement?

Answer

The patient can splint the stoma area with a hand or pillow during coughing, transfers, or resisted trunk activity to reduce strain and protect the peristomal skin and appliance seal.

#363 · Lymphatic Metabolic Endocrine GI and GU Systems

What ostomy appliance issue should prompt a physical therapist to modify or stop an exercise session?

Answer

Leakage of stool or urine from the appliance seal, peristomal skin breakdown, or a bulging stoma suggestive of parastomal hernia should prompt stopping exercise and addressing appliance fit.

#364 · Lymphatic Metabolic Endocrine GI and GU Systems

Why should abdominal binders be used cautiously during exercise in a patient with a recent ostomy?

Answer

An improperly placed binder can compress the stoma, impairing blood flow to the bowel or obstructing output, so binders should avoid direct pressure over the stoma site.

#365 · Lymphatic Metabolic Endocrine GI and GU Systems

What position is generally avoided early after abdominal or pelvic surgery due to increased intra-abdominal pressure?

Answer

Prolonged Valsalva maneuver, such as breath holding during resistance exercise, is avoided because it sharply raises intra-abdominal pressure and can stress the healing incision.

#366 · Lymphatic Metabolic Endocrine GI and GU Systems

What is stress urinary incontinence and what is its primary underlying cause?

Answer

Involuntary leakage of urine with activities that increase intra-abdominal pressure such as coughing, sneezing, or jumping. It is usually caused by pelvic floor muscle weakness or urethral hypermobility/sphincter insufficiency.

#367 · Lymphatic Metabolic Endocrine GI and GU Systems

What is urge urinary incontinence and what is its primary underlying mechanism?

Answer

Involuntary leakage accompanied by or immediately preceded by a strong, sudden urge to void. It results from involuntary detrusor muscle contractions, often called overactive bladder.

#368 · Lymphatic Metabolic Endocrine GI and GU Systems

What is mixed urinary incontinence?

Answer

A combination of stress and urge incontinence symptoms occurring in the same patient, requiring treatment strategies addressing both pelvic floor weakness and detrusor overactivity.

#369 · Lymphatic Metabolic Endocrine GI and GU Systems

What is overflow incontinence and what commonly causes it?

Answer

Frequent or constant dribbling of urine due to an overdistended bladder that cannot empty completely, commonly from bladder outlet obstruction (such as prostatic enlargement) or an underactive detrusor muscle.

#370 · Lymphatic Metabolic Endocrine GI and GU Systems

What is functional incontinence?

Answer

Urinary leakage that occurs because a person cannot reach the toilet in time due to physical or cognitive impairments, even though the urinary system itself functions normally.

#371 · Lymphatic Metabolic Endocrine GI and GU Systems

What is the primary first-line conservative treatment for stress urinary incontinence?

Answer

Pelvic floor muscle training (Kegel exercises) to strengthen the levator ani and improve urethral support and closure pressure during increases in abdominal pressure.

#372 · Lymphatic Metabolic Endocrine GI and GU Systems

What behavioral strategy is commonly used to treat urge incontinence and overactive bladder?

Answer

Bladder training with scheduled, progressively lengthened voiding intervals, combined with urge suppression techniques such as pelvic floor contraction (a 'quick flick') and distraction to inhibit the urge.

#373 · Lymphatic Metabolic Endocrine GI and GU Systems

How should a patient correctly perform a pelvic floor muscle contraction for a Kegel exercise?

Answer

Contract the muscles as if stopping the flow of urine or preventing passage of gas, lifting the pelvic floor upward and inward, without contracting the gluteal, abdominal, or hip adductor muscles as substitutions.

#374 · Lymphatic Metabolic Endocrine GI and GU Systems

What is biofeedback used for in pelvic floor rehabilitation?

Answer

It provides visual or auditory feedback of pelvic floor muscle activity, typically via surface EMG or manometry, to help patients learn correct contraction and relaxation patterns and avoid substitution strategies.

#375 · Lymphatic Metabolic Endocrine GI and GU Systems

What type of urinary incontinence is most common in men following radical prostatectomy?

Answer

Stress urinary incontinence, resulting from damage to the urethral sphincter mechanism and surrounding support structures during removal of the prostate.

#376 · Lymphatic Metabolic Endocrine GI and GU Systems

When is pelvic floor muscle training typically initiated after radical prostatectomy?

Answer

Often preoperatively when possible, and postoperatively once the urinary catheter is removed, since early pelvic floor training has been shown to speed return of continence.

#377 · Lymphatic Metabolic Endocrine GI and GU Systems

What erectile and continence-related complication is common after radical prostatectomy due to shared anatomy?

Answer

Erectile dysfunction can occur alongside stress incontinence because the neurovascular bundles controlling erectile function run close to the prostate and urethral sphincter and may be affected during surgery.

#378 · Lymphatic Metabolic Endocrine GI and GU Systems

What activity precaution is often given in the first weeks after prostatectomy regarding lifting and straining?

Answer

Avoid heavy lifting, straining, and high-impact activity for several weeks to protect the surgical site and reduce intra-abdominal pressure that could worsen leakage or delay healing.

#379 · Lymphatic Metabolic Endocrine GI and GU Systems

What is the difference between pelvic floor muscle overactivity and underactivity in terms of clinical presentation?

Answer

Overactivity presents with an inability to relax the pelvic floor, often causing pelvic pain, constipation, or urinary hesitancy, while underactivity presents with weakness contributing to incontinence and pelvic organ prolapse.

#380 · Lymphatic Metabolic Endocrine GI and GU Systems

What lifestyle factor is commonly addressed as a bladder irritant in patients with urge incontinence?

Answer

Caffeine intake is a common bladder irritant that can increase urinary urgency and frequency, so reducing caffeinated beverages is a standard behavioral recommendation alongside bladder training.

#381 · Integumentary System

What defines a Stage 1 pressure injury?

Answer

Intact skin with non-blanchable erythema over a bony prominence. Skin temperature, edema, or firmness may differ from surrounding tissue, but there is no open wound.

#382 · Integumentary System

What defines a Stage 2 pressure injury?

Answer

Partial-thickness skin loss with exposed dermis, presenting as a shallow open ulcer with a pink or red wound bed, or an intact or ruptured serum-filled blister. No slough or granulation tissue is present.

#383 · Integumentary System

What defines a Stage 3 pressure injury?

Answer

Full-thickness skin loss in which adipose tissue is visible in the ulcer, and granulation tissue and rolled wound edges are often present. Bone, tendon, and muscle are not exposed.

#384 · Integumentary System

What defines a Stage 4 pressure injury?

Answer

Full-thickness skin and tissue loss with exposed or directly palpable fascia, muscle, tendon, ligament, cartilage, or bone. Slough or eschar may be present, and undermining or tunneling is common.

#385 · Integumentary System

How is an unstageable pressure injury defined and staged?

Answer

Full-thickness skin and tissue loss in which the extent of tissue damage cannot be confirmed because the wound bed is obscured by slough or eschar. It cannot be staged until enough slough or eschar is removed to expose the base.

#386 · Integumentary System

What is a deep tissue pressure injury (DTPI)?

Answer

Intact or non-intact skin with a localized area of persistent non-blanchable deep red, maroon, or purple discoloration, or an epidermal separation revealing a dark wound bed or blood-filled blister, resulting from intense and/or prolonged pressure and shear at the bone-muscle interface.

#387 · Integumentary System

Why can a deep tissue injury not simply be called a Stage 4 wound on first assessment?

Answer

The true extent of underlying tissue damage is not yet visible at the surface, so staging is deferred until the wound evolves and demarcates, which may reveal a much deeper injury than initially apparent.

#388 · Integumentary System

A therapist finds a pressure injury covered entirely in thick black eschar. How should this be documented?

Answer

As an unstageable pressure injury, since the wound bed is obscured and the true depth cannot be determined until debridement occurs.

#389 · Integumentary System

What percentage of total body surface area (TBSA) does each arm represent using the Rule of Nines in an adult?

Answer

9% per arm (front and back combined), for a total of 18% for both upper extremities.

#390 · Integumentary System

What percentage of TBSA does each leg represent using the adult Rule of Nines?

Answer

18% per leg (front and back combined), for a total of 36% for both lower extremities.

#391 • Integumentary System

What percentage of TBSA do the head and the anterior and posterior trunk each represent in the adult Rule of Nines?

Answer

The head represents 9%, the anterior trunk 18%, and the posterior trunk 18%. Combined with 18% for both arms, 36% for both legs, and 1% for the genital region, these figures sum to 100%.

#392 • Integumentary System

How does the Rule of Nines differ for infants and young children compared to adults?

Answer

The head accounts for a larger proportion (about 18%) and the legs a smaller proportion (about 14% each) because an infant's head is proportionally larger relative to body size, so the Lund and Browder chart is used for more accurate pediatric estimation.

#393 • Integumentary System

Describe the clinical presentation of a superficial (first-degree) burn.

Answer

Involves only the epidermis, presenting as red, dry, painful skin without blisters, similar to a mild sunburn, and typically heals within 3 to 7 days without scarring.

#394 • Integumentary System

Describe the clinical presentation of a superficial partial-thickness (second-degree) burn.

Answer

Involves the epidermis and upper dermis, presenting as red, moist, blistered skin that is very painful and blanches with pressure, and typically heals in 1 to 3 weeks with minimal scarring.

#395 • Integumentary System

Describe the clinical presentation of a deep partial-thickness burn.

Answer

Involves the epidermis and deeper dermis, presenting as red or waxy white, wet or dry skin with decreased sensation and diminished capillary refill, requiring several weeks to heal and often needing grafting, with a high risk of hypertrophic scarring.

#396 • Integumentary System

Describe the clinical presentation of a full-thickness (third-degree) burn.

Answer

Destroys the epidermis and entire dermis, presenting as dry, leathery, white, brown, or charred skin that is insensate to pain because nerve endings are destroyed, and it requires surgical grafting to heal.

#397 · Integumentary System

What defines a fourth-degree burn?

Answer

A burn that extends through the entire skin into subcutaneous fat, muscle, tendon, or bone, resulting in charred, insensate tissue that requires extensive surgical management and often amputation or reconstruction.

#398 · Integumentary System

Why is a deep partial-thickness burn often more painful initially than a full-thickness burn, yet the pain can be misleading?

Answer

Because some nerve endings remain intact in the deeper dermis of a partial-thickness burn, whereas full-thickness burns destroy nerve endings entirely, so an insensate wound is actually a sign of greater severity, not less.

#399 · Integumentary System

Where do arterial (ischemic) ulcers typically occur, and what do they look like?

Answer

They occur distally, such as over the toes, lateral malleolus, and dorsum of the foot, presenting as small, round, deep ulcers with a pale, dry wound bed, minimal exudate, and well-defined punched-out borders.

#400 · Integumentary System

What pain pattern is characteristic of arterial ulcers, and why?

Answer

Pain increases with leg elevation and decreases with the leg in a dependent position, because elevation further reduces already compromised arterial blood flow while dependency uses gravity to assist perfusion.

#401 · Integumentary System

Where do venous insufficiency ulcers typically occur, and what do they look like?

Answer

They occur most often over the medial malleolus and distal leg, presenting as shallow, irregularly shaped ulcers with a ruddy red or yellow fibrous wound bed, heavy exudate, and surrounding hemosiderin staining and edema.

#402 · Integumentary System

What is the primary conservative treatment for venous insufficiency ulcers, and why does it work?

Answer

Compression therapy, because it counteracts venous hypertension and reduces edema by supporting venous return from the lower extremity back toward the heart.

#403 · Integumentary System

Why is compression therapy contraindicated in a patient with significant arterial insufficiency?

Answer

Compression can further restrict already compromised arterial blood flow, risking tissue ischemia and necrosis, so ankle-brachial index should be checked before compression is applied.

#404 · Integumentary System

Where do neuropathic (diabetic) ulcers typically occur, and what do they look like?

Answer

They occur over pressure and friction points on the plantar surface of the foot, such as under the metatarsal heads or the heel, presenting as deep ulcers often surrounded by callus, with a well-defined border and typically painless despite significant depth.

#405 · Integumentary System

Why are neuropathic ulcers often painless despite being deep, and what does this imply for treatment?

Answer

Peripheral sensory neuropathy reduces protective sensation, so the patient cannot feel the injury forming, which means offloading pressure with total contact casting or specialized footwear is essential since pain cannot serve as a warning sign.

#406 · System Interactions

A patient reports saddle anesthesia, bilateral leg weakness, and new urinary retention during a lumbar spine evaluation. What condition must be ruled out immediately?

Answer

Cauda equina syndrome. This is a surgical emergency requiring immediate physician referral because delayed decompression can cause permanent bowel, bladder, and lower extremity dysfunction.

#407 · System Interactions

How can a therapist distinguish cardiac chest pain from musculoskeletal chest pain during an exercise session?

Answer

Cardiac chest pain typically is not reproduced by palpation or a specific movement, often worsens with exertion and improves with rest, and may be accompanied by diaphoresis, nausea, or radiating arm/jaw pain. Musculoskeletal pain is usually reproducible with palpation or a specific motion.

#408 · System Interactions

List the classic clinical signs suggestive of a deep vein thrombosis (DVT) in the calf.

Answer

Unilateral calf swelling, warmth, redness, and tenderness. A positive Homans sign is classically taught but is now considered an outdated, unreliable screening tool and should never be used alone to rule DVT in or out. Any suspected DVT warrants immediate physician referral before continuing exercise or manual therapy.

#409 · System Interactions

Why should a therapist avoid aggressive calf palpation or stretching in a patient with suspected DVT?

Answer

Vigorous mobilization of a clot-containing vessel can dislodge the thrombus, risking pulmonary embolism. The patient should be referred for imaging (such as Doppler ultrasound) before any manual intervention proceeds.

#410 · System Interactions

What are common cancer red flags a physical therapist should screen for in a patient with unexplained back pain?

Answer

Unexplained weight loss, night pain that is unrelieved by rest or position change, history of cancer, age over 50, and constant progressive pain not linked to mechanical movement. These findings warrant physician referral.

#411 · System Interactions

A patient presents with pulsatile abdominal mass and back pain that is unrelated to movement. What vascular emergency should be suspected?

Answer

Abdominal aortic aneurysm (AAA). This requires immediate emergency referral, as rupture is life-threatening; palpation should be gentle and exercise should be withheld pending medical evaluation.

#412 · System Interactions

Why do beta blockers blunt the expected heart rate response to exercise, and what should the therapist use instead to monitor intensity?

Answer

Beta blockers reduce sympathetic stimulation of the heart, lowering resting and exercise heart rate and capping maximal heart rate. Therapists should use the Rating of Perceived Exertion (RPE) scale rather than target heart rate zones to gauge intensity.

#413 · System Interactions

What tendon-related risk is associated with long-term or repeated corticosteroid use, and which tendon is most commonly affected?

Answer

Corticosteroids weaken collagen structure and increase the risk of tendon rupture, most notably in the Achilles tendon. Therapists should modify loading and avoid aggressive stretching or plyometrics in patients on chronic steroid therapy.

#414 · System Interactions

A patient taking warfarin reports easy bruising and occasional nosebleeds. What precautions should the therapist take during manual therapy?

Answer

Use lighter-grade soft tissue and joint mobilization techniques, avoid deep tissue work or vigorous massage, and monitor for excessive bruising or bleeding, since anticoagulants increase bleeding and fall-related injury risk.

#415 · System Interactions

Why are patients on anticoagulant therapy at increased fall-related risk during balance training?

Answer

Even a minor fall can cause significant internal bleeding or large hematomas because clotting is impaired. Balance activities should be closely guarded and progressed cautiously with attention to environmental safety.

#416 · System Interactions

What medication class commonly causes orthostatic hypotension, and what precaution should a therapist take before mobilizing these patients?

Answer

Antihypertensives (especially diuretics, alpha blockers, and some calcium channel blockers) commonly cause orthostatic hypotension. The therapist should have the patient rise slowly, monitor blood pressure in supine, sitting, and standing positions, and guard closely during transfers.

#417 · System Interactions

A diabetic patient reports numbness in a stocking-glove distribution in the feet. What comorbidity consideration affects gait training and footwear education?

Answer

Peripheral neuropathy increases fall risk and risk of unnoticed skin breakdown or ulceration. Daily foot inspection, proper footwear, and careful sensory-based balance and gait training should be emphasized.

#418 · System Interactions

What is a key sign that chest pain during exercise may represent angina rather than a musculoskeletal strain?

Answer

Angina is typically described as pressure, tightness, or squeezing that is exertion-related, resolves with rest or nitroglycerin, and may radiate to the jaw, neck, or left arm, without reproduction by palpation.

#419 · System Interactions

When should a physical therapist refer a patient back to their physician regarding new-onset symptoms during a plan of care?

Answer

Whenever red flag symptoms emerge (unexplained weight loss, night pain, bowel/bladder changes, fever, progressive neurological deficits, or symptoms outside the expected pattern for the diagnosis) that fall outside the scope of physical therapy practice or suggest a change in medical status.

#420 · System Interactions

What is the significance of bilateral lower extremity edema combined with shortness of breath and fatigue in a patient with a cardiac history?

Answer

This combination suggests possible congestive heart failure exacerbation. The therapist should hold strenuous exercise, monitor vitals, and refer the patient for medical evaluation.

#421 · System Interactions

Why must a therapist use caution with resistance training in a patient on long-term oral corticosteroids?

Answer

Corticosteroids can cause muscle atrophy (steroid myopathy), osteoporosis, and tendon weakening, so loading should be progressed conservatively and high-impact or ballistic activities minimized.

#422 · System Interactions

A patient describes progressively worsening low back pain that is worse at night and unrelieved by any position. What screening category does this fall under and what is the appropriate action?

Answer

This is a red flag for possible neoplastic or systemic disease. The therapist should discontinue mechanical treatment approaches and refer the patient to their physician for further workup.

#423 · System Interactions

What effect do NSAIDs have that is relevant to a therapist treating a patient after a musculoskeletal injury or surgery?

Answer

NSAIDs can mask pain and inflammation, potentially allowing overexertion of healing tissue, and long-term use carries risk of gastrointestinal bleeding and renal impairment, so therapists should ask about GI symptoms and monitor tissue tolerance closely.

#424 · System Interactions

How should a therapist screen for cauda equina syndrome during a lumbar spine subjective history?

Answer

Ask specifically about saddle anesthesia, changes in bowel or bladder control (retention or incontinence), and bilateral leg symptoms. Any positive findings require same-day physician or emergency referral.

#425 · System Interactions

What vital sign pattern combined with unilateral leg swelling should raise suspicion for pulmonary embolism secondary to DVT?

Answer

Sudden onset of dyspnea, tachycardia, and pleuritic chest pain in a patient with known or suspected DVT is a medical emergency requiring immediate emergency referral, not continued physical therapy treatment.

#426 · System Interactions

Why is it important for a therapist to know a patient's full medication list before beginning an exercise program?

Answer

Medications can alter exercise response (beta blockers blunting heart rate), increase bleeding risk (anticoagulants), affect balance (sedatives, antihypertensives), or weaken connective tissue (corticosteroids), all of which change how exercise intensity and manual therapy should be modified.

#427 · System Interactions

A patient with a history of prostate cancer reports new unrelenting thoracic back pain. What screening consideration is critical here?

Answer

Metastatic bone disease should be suspected given the cancer history and constant, non-mechanical pain pattern. The therapist should refer back to the physician rather than proceeding with standard mechanical treatment.

#428 · System Interactions

What is the therapist's responsibility when a patient exhibits signs and symptoms outside their expected diagnosis or plan of care?

Answer

The therapist must recognize findings inconsistent with the working diagnosis, perform a systems review, and refer the patient to the appropriate medical provider, since physical therapists are trained to screen for medical conditions requiring referral, not to diagnose them.

#429 · System Interactions

Why might a patient on beta blockers report a high RPE at a heart rate that still seems well below target training zones?

Answer

Because beta blockers blunt the chronotropic response, the heart rate underrepresents true exertion, so RPE becomes a more accurate proxy for exercise intensity than heart rate-based formulas.

#430 · System Interactions

What abdominal symptom combination in an older adult during a lumbar or hip evaluation should prompt suspicion of an abdominal aortic aneurysm rather than a musculoskeletal source?

Answer

Deep, boring abdominal or back pain unrelated to movement or position, sometimes with a palpable pulsatile mass, in a patient with vascular risk factors such as smoking or hypertension, warrants urgent medical referral rather than continued musculoskeletal evaluation.

#431 · Equipment Devices and Therapeutic Modalities

Which assistive device provides the most stability for a patient with poor balance and bilateral lower extremity weakness?

Answer

A standard walker (or a rolling walker with a seat, for patients who need to rest) provides the widest base of support and greatest stability of the common assistive devices, more than axillary crutches, forearm crutches, or a cane.

#432 · Equipment Devices and Therapeutic Modalities

How should a standard cane be measured and on which side is it used?

Answer

The cane top should reach the greater trochanter with the elbow flexed about 20 to 30 degrees, and it is held in the hand opposite the involved lower extremity so the base of support widens and the reciprocal gait pattern is preserved.

#433 · Equipment Devices and Therapeutic Modalities

Describe the three-point gait pattern and who it is appropriate for.

Answer

Both assistive device tips (crutches or walker) and the involved limb advance together, followed by the uninvolved limb; it is used for patients who must remain fully non-weightbearing or partial weightbearing on one lower extremity.

#434 · Equipment Devices and Therapeutic Modalities

What is a four-point gait pattern and which patients benefit from it?

Answer

One assistive device tip or crutch advances, then the opposite lower extremity, then the other device tip, then the remaining leg, so three points always contact the ground; it is a very stable pattern used for patients with poor balance or bilateral involvement who can bear weight on both legs.

#435 · Equipment Devices and Therapeutic Modalities

How should axillary crutches be fitted for correct length and hand grip height?

Answer

The crutch tip sits about 2 inches lateral and 6 inches anterior to the foot with 2 to 3 finger widths of clearance between the axilla and the crutch pad, and the hand grip is set so the elbow is flexed about 20 to 30 degrees.

#436 · Equipment Devices and Therapeutic Modalities

Why are forearm (Lofstrand) crutches often preferred over axillary crutches for long-term ambulators?

Answer

The forearm cuff frees the hands for tasks without dropping the crutch and avoids pressure on the axillary vessels and radial nerve, reducing the risk of crutch palsy seen with prolonged axillary crutch use.

#437 · Equipment Devices and Therapeutic Modalities

What seat depth and height guidelines are used when prescribing a standard wheelchair?

Answer

Seat depth should leave about 1 to 2 finger widths of clearance behind the popliteal fossa, and seat height should allow about 2 inches of clearance between the popliteal fossa and the front edge of the seat while keeping the hips, knees, and ankles at roughly 90 degrees.

#438 · Equipment Devices and Therapeutic Modalities

What is the purpose of a solid seat insert or firm cushion base in wheelchair seating?

Answer

It prevents hammocking of a sling seat, which would otherwise pull the hips into adduction and internal rotation and promote poor pelvic and spinal alignment.

#439 · Equipment Devices and Therapeutic Modalities

Which wheelchair cushion type provides the best pressure redistribution for a patient at high risk of skin breakdown?

Answer

An air-filled (dynamic) cushion, such as a ROHO cushion, offers the greatest pressure redistribution by conforming to body contours and equalizing pressure over a larger surface area, compared to foam or gel cushions.

#440 · Equipment Devices and Therapeutic Modalities

What wheelchair modification is indicated for a patient with a below-knee amputation to maintain proper balance?

Answer

The rear axle is moved posteriorly (or amputee adapter brackets are added to move the rear wheels back) to shift the center of gravity backward and prevent the chair from tipping forward, since the residual limb and remaining leg weight are reduced.

#441 · Equipment Devices and Therapeutic Modalities

What is the difference between an ankle-foot orthosis (AFO) set in slight dorsiflexion versus one set in plantarflexion for a patient with dorsiflexor weakness?

Answer

A solid or hinged AFO set with a slight dorsiflexion assist prevents foot drop and toe drag during swing phase, while excessive plantarflexion set in the device can push the knee into hyperextension during stance.

#442 · Equipment Devices and Therapeutic Modalities

A patient ambulates with a foot slap gait deviation while wearing no orthosis. What is the most likely cause?

Answer

Weak dorsiflexors (commonly from a peroneal nerve injury) allow the foot to slap down uncontrolled after heel strike instead of lowering eccentrically; a solid or articulated AFO with dorsiflexion assist typically corrects this.

#443 · Equipment Devices and Therapeutic Modalities

What gait deviation results from a knee-ankle-foot orthosis (KAFO) locked in full extension, and why is it used?

Answer

It produces a circumducted or hip-hiked gait pattern to clear the foot during swing since the knee cannot flex; it is used for patients with quadriceps weakness or paralysis who need knee stability throughout stance.

#444 · Equipment Devices and Therapeutic Modalities

What is the primary difference in energy expenditure between a transtibial (below-knee) and transfemoral (above-knee) amputee during prosthetic gait?

Answer

Transfemoral amputees expend significantly more energy than transtibial amputees because they must control the prosthetic knee and have lost more of their own muscular leverage, and bilateral or higher-level amputations increase energy cost further.

#445 · Equipment Devices and Therapeutic Modalities

A transtibial prosthesis user shows lateral trunk bending toward the prosthetic side during stance. What is the likely cause?

Answer

The prosthesis may be too short, or there may be inadequate lateral wall support or hip abductor weakness; a prosthesis that is too long can instead cause vaulting, rising onto the toes of the sound limb, or circumduction, the prosthetic limb swinging outward during its own swing phase.

#446 · Equipment Devices and Therapeutic Modalities

What gait deviation results from excessive toe-out alignment of the prosthetic foot in a transtibial prosthesis?

Answer

It causes early knee flexion instability during stance because toe-out effectively shortens the lever arm anterior to the knee, reducing the extension moment that stabilizes the knee.

#447 · Equipment Devices and Therapeutic Modalities

What are the standard treatment parameters for NMES (neuromuscular electrical stimulation) used to strengthen a muscle?

Answer

A biphasic pulsed current with a pulse duration around 200 to 400 microseconds, frequency of 35 to 50 Hz, on:off ratio around 1:3 to 1:5, and intensity sufficient to produce a strong, visible tetanic contraction.

#448 · Equipment Devices and Therapeutic Modalities

What TENS parameters are used for conventional (high-rate) pain modulation versus low-rate (acupuncture-like) TENS?

Answer

Conventional TENS uses a high frequency (50 to 100+ Hz) with a short pulse duration and produces a sensory-level, non-painful tingling with rapid but short-lasting relief via the gate control mechanism; low-rate TENS uses a low frequency (1 to 5 Hz) with a longer pulse duration at a motor-level intensity, working through endorphin release with a slower onset but longer-lasting effect.

#449 · Equipment Devices and Therapeutic Modalities

List absolute contraindications for electrical stimulation.

Answer

A demand-type cardiac pacemaker or implanted defibrillator, application over the carotid sinus or anterior neck, over the pregnant uterus, and over areas of active cancer are all absolute contraindications to electrical stimulation.

#450 · Equipment Devices and Therapeutic Modalities

What is iontophoresis and what determines the polarity of the treatment electrode?

Answer

Iontophoresis uses a direct current to drive charged medication ions (such as dexamethasone for anti-inflammatory effect) through the skin; the treatment electrode must carry the same charge as the medication ion so it is repelled into the tissue, for example the negative electrode is used for dexamethasone.

#451 · Equipment Devices and Therapeutic Modalities

What is the difference between thermal and non-thermal (pulsed) therapeutic ultrasound, and what intensity/duty cycle distinguishes them?

Answer

Thermal ultrasound uses a continuous duty cycle (100 percent) at higher intensity to raise tissue temperature and increase extensibility of collagen, while non-thermal pulsed ultrasound uses a duty cycle of 20 percent or less to promote tissue healing effects like increased cell membrane permeability without significant heating.

#452 · Equipment Devices and Therapeutic Modalities

How does frequency selection affect the depth of penetration in therapeutic ultrasound?

Answer

A lower frequency (around 1 MHz) penetrates deeper (about 3 to 5 cm) and is used for deeper tissues, while a higher frequency (around 3 MHz) penetrates more superficially (about 1 to 2 cm) and concentrates energy closer to the skin surface.

#453 · Equipment Devices and Therapeutic Modalities

List absolute contraindications for therapeutic ultrasound.

Answer

Ultrasound should never be applied over the pregnant uterus, over the eyes, over a pacemaker or the thoracic region with a pacemaker present, over malignant tumors, over the epiphyseal plates in children, or directly over a healing fracture site, cemented prosthetic joints, or the gonads.

#454 · Equipment Devices and Therapeutic Modalities

What are the primary physiological effects of cryotherapy (cold application)?

Answer

Cryotherapy causes vasoconstriction, decreased nerve conduction velocity, decreased metabolic rate and enzymatic activity, decreased muscle spasm, and an initial analgesic effect, making it useful for acute inflammation and pain.

#455 · Equipment Devices and Therapeutic Modalities

What conditions are contraindications to applying cryotherapy?

Answer

Cryotherapy is contraindicated in patients with Raynaud's phenomenon, cold urticaria, cryoglobulinemia, paroxysmal cold hemoglobinuria, and impaired circulation or sensation, since it can cause tissue damage or an adverse vascular response in these conditions.

#456 · Equipment Devices and Therapeutic Modalities

What are the primary physiological effects of superficial heat (thermotherapy)?

Answer

Superficial heat causes vasodilation, increased blood flow and metabolic rate, increased tissue extensibility and collagen elasticity, decreased joint stiffness, and mild analgesia, making it useful for subacute or chronic conditions and before stretching.

#457 · Equipment Devices and Therapeutic Modalities

Why is superficial heat contraindicated over an area of acute inflammation or a recent injury?

Answer

Heat causes vasodilation and increased blood flow, which would increase edema, bleeding, and inflammation in an acute injury; cryotherapy is preferred in the first 24 to 72 hours to promote vasoconstriction instead.

#458 · Equipment Devices and Therapeutic Modalities

What is the general contraindication shared by both superficial heat and cold modalities?

Answer

Both are contraindicated over areas of impaired sensation or impaired circulation, over malignancy, and directly over the eyes, since the patient cannot reliably report a burn or frostbite reaction and the tissue response may be abnormal.

#459 · Equipment Devices and Therapeutic Modalities

What angle of pull and force are typically used for mechanical lumbar traction to produce mechanical separation of the vertebrae?

Answer

Lumbar traction typically uses a static or intermittent pull, with hips and knees flexed to flatten the lumbar lordosis. At least 25 percent of body weight overcomes trunk-table friction to produce a measurable effect, while roughly 50 percent of body weight or more is generally needed to achieve true mechanical separation of the vertebral segments.

#460 · Equipment Devices and Therapeutic Modalities

List contraindications to mechanical spinal traction.

Answer

Mechanical traction is contraindicated in acute spinal cord compression, spinal instability or fracture, severe osteoporosis, active malignancy in the spine, uncontrolled hypertension, and in acute inflammatory conditions such as rheumatoid arthritis affecting the cervical spine.

#461 · Safety Protection and Professional Responsibilities

What is the correct hand hygiene technique when hands are visibly soiled?

Answer

Wash with soap and water for at least 20 seconds. Alcohol-based hand sanitizer is not effective against visibly soiled hands or against spore-forming organisms such as *Clostridioides difficile*.

#462 · Safety Protection and Professional Responsibilities

What type of isolation precaution is required for a patient with active pulmonary tuberculosis?

Answer

Airborne precautions, which require a negative pressure room and an N95 respirator or higher-level mask for anyone entering the room.

#463 · Safety Protection and Professional Responsibilities

Which organisms require droplet precautions?

Answer

Organisms spread by large respiratory droplets such as influenza, pertussis, and meningococcal meningitis, requiring a surgical mask worn within about 3 feet (1 meter) of the patient.

#464 · Safety Protection and Professional Responsibilities

What precaution category applies to MRSA and VRE?

Answer

Contact precautions, requiring a gown and gloves for any physical contact with the patient or contaminated surfaces, with hand hygiene before and after glove removal.

#465 · Safety Protection and Professional Responsibilities

What is the correct order for donning personal protective equipment?

Answer

Gown first, then mask or respirator, then goggles or face shield, and gloves last, so gloves cover the gown cuffs.

#466 · Safety Protection and Professional Responsibilities

What is the correct order for doffing personal protective equipment?

Answer

Gloves first, then goggles or face shield, then gown, and mask or respirator last, with hand hygiene performed immediately after glove removal and again after all PPE is removed.

#467 · Safety Protection and Professional Responsibilities

Which standardized outcome measure is commonly used to assess fall risk in community-dwelling older adults?

Answer

The Berg Balance Scale and the Timed Up and Go test are both commonly used, with a Timed Up and Go score greater than 12 seconds associated with increased fall risk.

#468 · Safety Protection and Professional Responsibilities

What Timed Up and Go score is generally considered indicative of increased fall risk?

Answer

A time greater than 12 seconds is associated with increased fall risk in older adults.

#469 · Safety Protection and Professional Responsibilities

Name three intrinsic risk factors for falls in older adults.

Answer

Examples include impaired balance or gait, orthostatic hypotension, polypharmacy or sedative use, decreased visual acuity, and lower extremity weakness.

#470 · Safety Protection and Professional Responsibilities

Name three extrinsic (environmental) risk factors for falls.

Answer

Examples include poor lighting, loose rugs or clutter, lack of grab bars, uneven flooring, and inappropriate footwear.

#471 • Safety Protection and Professional Responsibilities

What is orthostatic hypotension and how is it defined?

Answer

A drop in systolic blood pressure of at least 20 mmHg or diastolic blood pressure of at least 10 mmHg within three minutes of standing from sitting or lying, which can cause dizziness and falls.

#472 • Safety Protection and Professional Responsibilities

What should a therapist do first if a patient becomes symptomatic with orthostatic hypotension during a transfer?

Answer

Lower the patient back down or return them to a seated or supine position immediately to restore cerebral perfusion and prevent a fall or loss of consciousness.

#473 · Safety Protection and Professional Responsibilities

What is the proper base of support width during a manual patient transfer to maximize stability?

Answer

The base of support should be widened to approximately shoulder width, with one foot slightly forward, to increase the stability of the therapist's stance.

#474 · Safety Protection and Professional Responsibilities

What body mechanics principle reduces spinal load when lifting a patient or object?

Answer

Keep the load close to the body's center of gravity, bend at the hips and knees rather than the spine, and avoid twisting the trunk during the lift.

#475 · Safety Protection and Professional Responsibilities

What is the recommended gait belt placement for a standing pivot transfer?

Answer

The gait belt should be placed snugly around the patient's waist over clothing, with room for two to four fingers underneath. The therapist grasps it with an underhand grip, which provides a secure hold without compromising the clinician's own grip strength if the patient begins to fall.

#476 · Safety Protection and Professional Responsibilities

During a sit to stand transfer, where should the therapist guard a patient with weakness on one side?

Answer

The therapist should guard from the affected (weaker) side to provide support and prevent the patient from falling toward that side.

#477 · Safety Protection and Professional Responsibilities

What is the first action in the Basic Life Support (BLS) adult chain of survival upon finding an unresponsive person?

Answer

Check for responsiveness and normal breathing, then call for help and activate the emergency response system (or send someone to call) before beginning chest compressions if there is no pulse.

#478 · Safety Protection and Professional Responsibilities

What is the recommended compression rate and depth for adult CPR?

Answer

100 to 120 compressions per minute at a depth of at least 2 inches (5 cm) but no more than 2.4 inches (6 cm), allowing full chest recoil between compressions.

#479 · Safety Protection and Professional Responsibilities

What is the compression to ventilation ratio for single-rescuer adult CPR?

Answer

30 compressions to 2 breaths for a single rescuer performing adult CPR without an advanced airway in place.

#480 · Safety Protection and Professional Responsibilities

What should a therapist do if an automated external defibrillator (AED) advises no shock?

Answer

Immediately resume chest compressions and continue CPR, rechecking the rhythm as prompted by the AED approximately every two minutes.

#481 • Safety Protection and Professional Responsibilities

What are the seven APTA Core Values for the physical therapy profession?

Answer

Accountability, Altruism, Compassion and Caring, Excellence, Professional Duty, Integrity, and Social Responsibility are the APTA Core Values that guide professional behavior.

#482 • Safety Protection and Professional Responsibilities

According to the APTA Code of Ethics, to whom does a physical therapist owe the primary duty of care?

Answer

The patient or client, whose interests must be placed above the therapist's own financial, institutional, or personal interests.

#483 · Safety Protection and Professional Responsibilities

What ethical principle requires a physical therapist to disclose a treatment error to the patient?

Answer

The principle of accountability and honesty requires the therapist to disclose any error that affects the patient's care in a timely and transparent manner.

#484 · Safety Protection and Professional Responsibilities

Under most state practice acts, can a Physical Therapist Assistant (PTA) perform an initial evaluation?

Answer

No. Evaluation, initial examination, and establishment of the plan of care are within the sole scope of the physical therapist and cannot be delegated to a PTA.

#485 · Safety Protection and Professional Responsibilities

What is required of a physical therapist regarding supervision of a PTA treating a patient?

Answer

The physical therapist must provide periodic reexamination and documented supervision, with the frequency of required on-site or direct supervision varying by state practice act.

#486 · Safety Protection and Professional Responsibilities

What does the acronym SOAP stand for in clinical documentation?

Answer

Subjective, Objective, Assessment, and Plan, a standardized format for organizing progress notes and treatment documentation.

#487 · Safety Protection and Professional Responsibilities

What type of information belongs in the Subjective section of a SOAP note?

Answer

The patient's own reported symptoms, pain level, functional complaints, and statements about their condition since the last visit.

#488 · Safety Protection and Professional Responsibilities

What type of information belongs in the Objective section of a SOAP note?

Answer

Measurable and observable data such as range of motion, manual muscle test grades, vital signs, gait distance, and specific interventions performed.

#489 · Safety Protection and Professional Responsibilities

Under HIPAA, what is considered protected health information (PHI)?

Answer

Any individually identifiable health information related to a patient's past, present, or future physical or mental health condition, treatment, or payment for care, transmitted or maintained in any form.

#490 · Safety Protection and Professional Responsibilities

What are the essential elements required for valid informed consent before initiating physical therapy treatment?

Answer

The patient must be given adequate information about the proposed intervention, its risks and benefits, and reasonable alternatives, must have decision-making capacity, and must voluntarily agree without coercion.

#491 • Research and Evidence-Based Practice

In the traditional hierarchy of evidence, what study design sits at the top as the strongest source of evidence for treatment effectiveness?

Answer

Systematic reviews and meta-analyses of randomized controlled trials sit at the top, since they pool results across multiple RCTs to produce the most precise and generalizable estimate of effect.

#492 • Research and Evidence-Based Practice

What is the defining feature of a randomized controlled trial (RCT) that distinguishes it from a cohort study?

Answer

In an RCT the researcher randomly assigns participants to intervention or control groups, which controls for confounding variables, while in a cohort study exposure or treatment is observed rather than assigned.

#493 · Research and Evidence-Based Practice

How does a case-control study differ from a cohort study in its basic design direction?

Answer

A case-control study starts with an outcome (cases with the condition versus controls without it) and looks backward for exposures, while a cohort study starts with an exposure and follows participants forward to see who develops the outcome.

#494 · Research and Evidence-Based Practice

What is a case series and what is its main limitation as a form of evidence?

Answer

A case series is a descriptive report of outcomes for a group of patients who received the same intervention, with no comparison group, so it cannot establish causation and is considered low-level evidence.

#495 • Research and Evidence-Based Practice

Define reliability as it applies to an outcome measure used in physical therapy.

Answer

Reliability is the consistency or reproducibility of a measurement, including test-retest reliability (same rater over time), intra-rater reliability (same rater repeated), and inter-rater reliability (different raters), often quantified with an intraclass correlation coefficient.

#496 • Research and Evidence-Based Practice

What is construct validity and how does it differ from criterion validity?

Answer

Construct validity is the extent to which a tool measures the theoretical construct it claims to measure, whereas criterion validity is the extent to which a tool's scores correlate with a recognized gold standard measure of the same construct.

#497 · Research and Evidence-Based Practice

What does the minimal clinically important difference (MCID) represent for an outcome measure?

Answer

The MCID is the smallest change in a measure's score that a patient would perceive as meaningful or beneficial, distinguishing a clinically important change from one that is merely statistically significant.

#498 · Research and Evidence-Based Practice

A special test has high sensitivity but low specificity. What does this mean for its clinical use?

Answer

A highly sensitive test has few false negatives, so a negative result helps rule out the condition (mnemonic SnNout), but its low specificity means a positive result is not very useful for ruling the condition in due to a high false positive rate.

#499 · Research and Evidence-Based Practice

How is a positive likelihood ratio (LR+) interpreted, and what does an LR+ greater than 10 indicate?

Answer

The LR+ tells how much a positive test result increases the odds of the target condition being present, and an LR+ greater than 10 represents a large and often conclusive shift toward ruling the condition in.

#500 · Research and Evidence-Based Practice

In research statistics, what does a 95% confidence interval that does not cross zero (for a mean difference) tell you about statistical significance?

Answer

It indicates the result is statistically significant at the p less than 0.05 level, because a confidence interval excluding the null value of zero shows the true effect is unlikely to be no difference at all.